

The label economics of calibration transfer: a five-benchmark, dual-modality evaluation of simple corrections versus deep and physics-informed representations

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ABSTRACT

Near-infrared calibration models degrade sharply across instruments, seasons and origins, yet when an expensive deep or physics-informed representation is worth deploying rather than the simplest correction has not been evaluated systematically under controlled target-label budgets. Under one unified protocol we evaluate five benchmarks spanning four shift types and two modalities (corn and IDRC tablets across instruments, OSSL soil mid-infrared across laboratories, mango across seasons, in-house apples across origins/years): 126 transfer tasks, five fixed seeds, and target-clustered bootstrap inference. (1) At target-label budgets ≤ 5 –10, label-free second-order moment alignment (CORAL) and a two-parameter slope/bias correction match or exceed deep and physics-informed representations: clearly better on the instrument and seasonal benchmarks, comparable on cross-origin apple, and indistinguishable on cross-laboratory soil. A pre-specified selection-free reanalysis preserves this and reverses an apparent near-zero-label deep edge on apple. Because each shift type is carried by essentially one benchmark ($n = 5$), these are benchmark-level, not mechanism-level, effects. (2) Transductive CORAL raises the benchmark-median performance-to-deviation ratio from 0.61 to 2.01 under instrument shift, though only about 53% of the 30 tasks individually reach $\text{RPD} \geq 2$. (3) The most frequent best method tracks benchmark $\times$ label budget, reported as an in-sample table requiring prospective validation, not a decision rule. (4) The crossover budget n^* is not predicted by continuous shift measures (maximum mean discrepancy, Wasserstein distance, intrinsic dimensionality; $R^2 \leq 0.25$); with $n = 5$ and shift near-collinear with dataset, this is a bounded empirical observation, not an existence claim.

1 Introduction

Near-infrared (NIR) and mid-infrared (MIR) spectroscopy combined with chemometric modelling has become a mature technique for rapid, non-destructive quantification in agricultural produce quality, soil properties and pharmaceutical quality control[1, 2]. Under the controlled conditions of a single instrument and a single batch, models such as partial least squares regression (PLSR) frequently exceed a ratio of performance to deviation (RPD) of 2.0, which meets application requirements[3, 4]. Such calibration models, however, generally lose accuracy abruptly when deployed across instruments, batches, seasons, growing origins or laboratories: when a model calibrated on one instrument or at one origin is used to predict samples acquired under another condition, the root mean square error

often increases several fold[5, 6]. Calibration transfer—bridging the systematic spectral distribution shift between a source domain and a target domain while using as few target-domain calibration labels as possible, or none at all—has therefore become a bottleneck common to the large-scale deployment of quantitative spectroscopy, spanning grain, tablets, soil and fruit. For apple soluble solids content (SSC, i.e. sugar content), our group has previously established the effectiveness of visible/near-infrared spectroscopy combined with functional linear regression and fractional-differential ensemble learning[7–11]; performance nonetheless degrades substantially under cross-origin deployment, and this is the concrete setting used here as an in-depth case study.

Around this bottleneck the methodological landscape has expanded steadily, and falls broadly into four families of increasing complexity and cost. The first is spectral preprocessing—multiplicative scatter correction, standard normal variate transformation and

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derivative transforms—which partially normalises scattering differences[12–15]. The second is empirical calibration transfer, such as direct standardisation, piecewise direct standardisation (PDS)[16], slope/bias correction and second-order moment alignment (correlation alignment, CORAL)[17], together with their recent variants[18–23]; this family is markedly effective against inter-instrument differences and batch drift, but mostly relies on paired target-domain samples or a small number of target calibration samples[24–27]. The third is statistical and deep domain adaptation, which achieves (nearly) unsupervised adaptation by aligning source and target feature distributions; representative methods include domain-adversarial neural networks (DANN)[28], deep second-order moment alignment (Deep CORAL)[29], maximum mean discrepancy (MMD) regularisation[30] and one-dimensional convolutional networks[31–41]. The fourth is physics-informed machine learning, which encodes priors such as conservation laws and the Beer–Lambert (BL) law into the network in the hope of improving robustness in data-scarce and extrapolative regimes[42–46].

Methods have grown steadily more complex and more expensive, yet one question of first-order practical importance has never been answered systematically: given a concrete calibration transfer and an affordable target-domain labelling budget, when is it worth deploying an expensive deep or physics-informed representation, and when does the simplest classical correction already suffice? Existing evaluations are almost always conducted on a single dataset and a single shift type, and commonly exhibit a tendency to report only the victories of complex methods while under-representing simple baselines; a unified head-to-head comparison across benchmarks, across modalities and under controlled label budgets is lacking. Beyond this, whether a “transferability scaling law” exists—one that could predict, from measurable quantities of domain shift (distribution distances, intrinsic dimensionality), the crossover label budget n^* at which a simple method catches up with a complex one—has never been genuinely tested; if such a law held, it would provide an a priori basis for method selection, and if it did not, that would show that transfer difficulty cannot be characterised by a single scalar.

Under a unified protocol, this study examines five calibration-transfer benchmarks spanning four types of domain shift and two spectral modalities—corn across three instruments[47], tablets across two instruments (IDRC 2002 shoot-out)[48], global soil mid-infrared across laboratories (Open Soil Spectral Library, OSSL)[49], mango across seasons[50], and in-house apples across origins and years—comprising 126 transfer tasks, and, with five fixed random seeds and bootstrap inference clustered on the target set, provides a controlled empirical comparison of three families of methods (simple classical corrections, classical model

updating, and deep/physics representations) at each label budget. The objectives are: (1) to establish a label-budget reckoning of “simple correction vs deep/physics representation” and to determine the method-selection boundaries under each type of shift; (2) to quantify the feasibility of label-free second-order moment alignment and to organise “benchmark (shift label) $\times$ label budget $\rightarrow$ most frequent best method” into an in-sample frequency table (a heuristic map requiring prospective validation); and (3) to test whether the crossover budget n^* can be predicted from measures of domain shift, that is, whether a “unified transferability scaling law” holds. Beyond this unified reckoning, cross-origin transfer on our in-house apple data serves as an in-depth case study, in which we test whether a physics-informed factorisation architecture that encodes the BL law as an inductive bias in latent space (Phys+BL) can outperform classical calibration transfer, and delimit its scope of application honestly. The stance of this paper is that of an honest multi-benchmark reckoning: we do not presuppose that complex methods are better, we report faithfully when simple baselines suffice and where they are indistinguishable from complex ones, and we point out the incommensurability of domain shift across mechanisms.

2 Materials and methods

2.1 Five calibration-transfer benchmarks and a unified transfer-task matrix

To compare method selection across benchmarks and modalities, we assembled five public or in-house calibration-transfer benchmarks covering four types of domain shift and two spectral modalities (Table 1). (1) **Corn**[47]: 80 corn samples acquired on three NIR instruments (m5, mp5, mp6), 1100–2498 nm (700 channels), with moisture, oil, protein and starch contents as the four label attributes; differences between instruments constitute *instrument-response shift*. (2) **Tablet**[48]: 655 tablets from the IDRC 2002 shoot-out acquired on two transmission NIR instruments, 600–1898 nm (650 channels), with tablet weight, hardness and active-ingredient assay as the three label attributes; also *instrument shift*. (3) **Soil**[49]: the mid-infrared (MIR) subset of the OSSL global soil spectral library, taking the KSSL, AFSIS1 and CAF sub-libraries established by different laboratories; the raw high-dimensional absorbance spectra were binned at equal wavenumber intervals to 340 dimensions, with total carbon and sand content as labels; differences between sub-libraries constitute *cross-laboratory shift*, and this is the only MIR modality in this study. (4) **Mango**[50]: the visible–near-infrared spectra of intact mango fruit from Anderson et al. across four harvest seasons (binned to 103 dimensions), with dry matter (DM) content as the label; differences between seasons

Table 1

Composition of the five calibration-transfer benchmarks and the transfer-task matrix

Benchmark	Shift type	Modality	Domains (n)	Bands	Label attributes	Tasks
Corn	Instrument	NIR	3 instruments	700	moisture/oil/protein/starch	24
Tablet	Instrument	NIR	2 instruments	650	weight/hardness/assay	6
Mango	Season	NIR	4 harvest seasons	103	dry matter	12
Soil (OSSL)	Laboratory	MIR	3 sub-libraries	340	total carbon/sand	12
Apple	Origin/year	NIR	origin–year–instrument	229	SSC	72
Total	4 types	2 modalities	—	—	—	126

Note: “Domains (n)” is the number of atomic acquisition conditions within the benchmark; “Tasks” is the number of combinations of ordered domain pairs with label attributes. Corn, tablet and soil are public benchmarks; mango is the public dataset of Anderson et al.; apple was acquired in house (Section 2.4). Soil is the only mid-infrared (MIR) modality.

constitute *seasonal shift*. (5) **Apple** (in-house): diffuse-reflectance spectra of intact ‘Fuji’ apples acquired in 2018, 2019 and 2025 across three origins per year and two instrument generations (aligned to 229 common wavelength channels), with SSC as the label. The three years $\times$ three origins give nine atomic origin–year domains, whose ordered pairs across all attributes yield the 72 apple transfer tasks; the 2019 season (instrument A, acquired and cleaned as for 2018) enters the unified benchmark but not the finer case study of Sections 2.4–3.7, which contrasts the 2018 and 2025 seasons. The compound differences of origin $\times$ year $\times$ instrument constitute *cross-origin/year shift*. Acquisition and cleaning of the apple data are detailed in Section 2.4.

Transfer tasks were formed by combining the ordered domain pairs within each benchmark with the label attributes: the source domain provides the calibration and the target domain is the condition to be transferred to. This yields 126 transfer tasks—corn 24 (6 ordered instrument pairs $\times$ 4 attributes), tablet 6 (2×3), mango 12 (4 seasons pairwise ordered $\times$ 1 attribute), soil 12 (6 ordered laboratory pairs $\times$ 2 attributes) and apple 72 (ordered multi-domain pairs $\times$ SSC)—which by shift type comprise 30 instrument, 72 cross-origin/year, 12 seasonal and 12 cross-laboratory tasks, and by modality 114 NIR and 12 MIR tasks. All benchmarks share the same evaluation pipeline: the full source domain trains the source model; the target domain is randomly partitioned into a held-out test set and a disjoint calibration pool, from which the n target calibration labels of each budget $n \in \{0, 5, 10, 20, 40\}$ are drawn. The same held-out test samples serve every budget and every method within a given draw—so differences along a budget curve reflect adaptation rather than a changing evaluation set, and the $n = 0$ methods (direct transfer and label-free CORAL) are scored on exactly those test samples—and the partition is redrawn and the task repeated independently under five fixed random seeds. The source, licence and SHA-256 checksum of every dataset are given in the data card (Table 2); the public benchmarks are not re-hosted here and should be obtained from their official sources.

2.2 Three families of methods

Three families of methods, of increasing complexity and cost, were compared under the same pipeline. **Simple classical corrections**: (i) second-order moment alignment, CORAL[17], which uses only unlabelled target spectra to align the covariance of source-domain features to that of the target domain and *uses no target labels whatsoever*; (ii) slope/bias correction (SBC), which corrects the scale and offset of the source model’s predictions by univariate linear regression on a small number of target-domain calibration samples, using only two parameters. **Classical model updating** (requiring more target labels): piecewise direct standardisation, PDS[16]; model updating (retraining after pooling source-domain and target-domain calibration samples); and target-only calibration (building a PLS model from scratch on target-domain calibration samples alone, without using the source domain). **Deep/physics representations**: a one-dimensional convolutional neural network (CNN); the domain-adversarial network DANN[28]; deep second-order moment alignment, Deep CORAL[29]; and the physics-informed BL factorisation architecture Phys+BL (Section 2.7). For fairness, all methods share the same source/target split and random seed within each task and operate on standard-normal-variate (SNV) standardised spectra, so that zero-label direct transfer of the source PLS model is itself a PLSR+SNV baseline; under the strict zero-label protocol, early stopping and weight selection for the deep methods instead use a source-domain hold-out set and touch no target labels. The implementation settings of every method are summarised in Table 3; note that this unified benchmark uses SNV standardisation, whereas the in-depth apple case study of Sections 2.4–2.7 is a separate pipeline with its own preprocessing (wavelet denoising and Savitzky–Golay smoothing, scatter correction deliberately omitted, Section 2.5), so absolute numbers are not comparable between the two.

Table 2

Data card: source, licence and traceability checksum for every dataset used in this study. The upper block lists the four public benchmarks (obtained from their official sources and not re-hosted here); the lower block lists the three in-house apple year-domains that make up the cross-origin benchmark.

Dataset	Source	Licence / availability	SHA-256 prefix (16 hex)
Corn (Cargill)[47]	Eigenvector Research	Free for research use ^a	e28fd4be274a54ca
Tablet (IDRC 2002 shoot-out)[48]	Eigenvector Research	Free for research use ^a	129a32ec9e194e56
Soil (OSSL)[49]	OpenGeoHub / Soil Spectroscopy	CC-BY 4.0	2044946ad5a10af0
Mango (Anderson)[50]	Mendeley Data	CC-BY 4.0	86159e2df9244fa5
Apple 2018, instrument A (in-house)	Xinjiang Agric. Univ.	Restricted ^b	a248445fd874bbf0
Apple 2019, instrument A (in-house)	Xinjiang Agric. Univ.	Restricted ^b	1495630ea432f4fb
Apple 2025, instrument B (in-house)	Xinjiang Agric. Univ.	Restricted ^b	bc6337e462fe5006

Note: the SHA-256 column is the first 16 hexadecimal characters of the SHA-256 checksum—of the raw source file for the public benchmarks, and of the cleaned per-year table (produced by the preprocessing script) for the apple data—and matches the values registered in the code repository's data cards. ^a Free for research use per the publisher's terms. ^b Internal data of Xinjiang Agricultural University, not publicly released; available from the corresponding author (huanghua@xjau.edu.cn) on reasonable request, subject to the university's data-sharing policy. Public benchmarks are not re-hosted; please download them from the official sources cited above and cite the original works.

Table 3

Implementation settings of the methods in the unified benchmark (all on SNV-standardised spectra, five fixed seeds). Deep methods run on CUDA with deterministic kernels; classical methods are CPU/scikit-learn. Full code is in the repository (Data availability).

Method	Architecture / key setting	Training	Method-specific
PLS (target-only, model update)	PLS regression	components 2–15, 3-fold CV	—
CORAL	second-moment alignment	closed form, transductive	covariance shrinkage $\varepsilon = 10^{-2}$
SBC (slope/bias)	affine $y = a\hat{y} + b$	least squares on n target labels	—
PDS	piecewise direct standardisation	per-channel Ridge ($\alpha = 1$)	window half-width 7; paired standards only
CNN (1D)	Conv 1→16→32 (k7), 64-d features, Linear 64→1	400 epochs, Adam, lr 10^{-3} , wd 10^{-4}	early stop patience 50 (source hold-out)
DANN	shared encoder + gradient-reversal domain head	150 epochs, Adam, lr 10^{-3} , wd 10^{-4}	GRL weight 0→0.5 schedule; no early stop
Deep CORAL	shared encoder, deep second-moment alignment	150 epochs, Adam, lr 10^{-3} , wd 10^{-4}	CORAL loss weight $\beta = 1.0$; no early stop
Phys+BL	$K = 8$ non-negative factorisation, 64-d features	500 epochs, Adam, lr 10^{-3} , wd 10^{-4}	reconstruction weight $\lambda = 0.3$; early stop patience 60

Note: the four deep methods share one convolutional backbone (Conv 1→16→32, kernel 7, adaptive average pooling, 64-dimensional features); fine-tuning at nonzero label budgets uses 80 epochs at lr 5×10^{-4} . PLS component number is selected within each fold by 3-fold cross-validation. This unified-benchmark backbone is distinct from the apple case-study network of Sections 2.7–2.8.

2.3 Label-budget reckoning, the crossover budget n^* and the unified statistical protocol

Each (task, method, label budget) cell was evaluated repeatedly under five fixed random seeds, and the across-seed median root mean square error (denoted RMSEP) was taken as the performance of that cell; usability was expressed as RPD = $\sigma_{y,\text{tgt}}/\text{RMSEP}$ (following common but application-dependent screening conventions[51], RPD ≥ 2 is treated as screening-level and ≥ 1.4 as preliminary). The core comparison is the task-wise paired difference between the *best simple method* (the better of CORAL and SBC) and the *best deep/physics method* (the best of CNN/DANN/Deep CORAL/Phys+BL) on each task: significance by the Wilcoxon signed-rank test (the distributions being

heavy-tailed and non-normal), effect size reported as Cliff's δ , confidence intervals from a cluster bootstrap *clustered on task (target set)* (≥ 2000 resamples), and multiple comparisons corrected by the Holm procedure within a pre-specified family of hypotheses. Cliff's δ is the unpaired dominance statistic—the fraction of all cross-family task pairs on which the simple family has the smaller error minus the fraction on which the deep family does, positive values favouring the simple family—and we report alongside it the paired per-task win fraction (“Simple-win %” in Tables 4 and 5), so that a δ of +1 is transparently the case of complete separation between the two error distributions and the win fraction its paired complement. Because taking the per-task best within each family is a selection that favours the larger (deep) family, the pre-specified

selection-free reanalysis (Table 5) rather than the best-of-family envelope is treated as the primary basis for the simple-versus-deep comparison.

The crossover budget n^* is defined, consistently throughout, as the minimum target-label budget on the grid $\{0, 5, 10, 20, 40\}$ at which the best simple method catches up with or overtakes the best deep/physics method; we additionally define n_{usable} as the minimum number of labels required to reach a given RPD threshold. Because n^* is read off a discrete budget grid it is interval-censored (and right-censored when no crossover occurs by 40 labels), so the log–log fits below are a deliberately conservative test of predictability rather than an exact continuous regression. To test the “transferability scaling law”, we regressed n^* on measures of domain shift in log–log space; the predictors comprise the maximum mean discrepancy (MMD) and the sliced Wasserstein distance between source and target distributions, the intrinsic dimensionality of the target domain (effective PCA dimension) and the signal-to-noise ratio. Extrapolability was tested by leave-one-benchmark-out (LOBO: fit on the remaining benchmarks, predict the held-out benchmark) together with conformal 90% prediction intervals. All formal experiments used the five fixed random seeds [20060515, 20041210, 19810915, 2023, 2024].

2.4 Apple case study: experimental material and spectral acquisition

As an in-depth case study alongside the unified reckoning, we analyse cross-origin transfer on our in-house apple data at finer granularity (results in Sections 3.4–3.7). Note that in the unified benchmark the apple data count as 72 transfer tasks (Section 2.1, comprising the three year-domains 2018, 2019 and 2025), whereas the case study below focuses on the 2018 and 2025 seasons and examines the physics-informed architecture over a finer set of 602 source→target scenarios; the two are complementary analyses on different terms. Two independent datasets were acquired, both of intact ‘Fuji’ apples at commercial maturity, in diffuse-reflectance mode, with reference SSC determined on the pressed juice by a digital refractometer (Atago PR-201α) at an acquisition temperature of $20 \pm 2^\circ\text{C}$.

The 2018 dataset comprises 630 samples with a single scan per fruit; the raw spectrometer range was 591–1001 nm with 1213 channels. For the 2025 dataset, five measurement facets per fruit were acquired, giving 3155 spectra in total, which were aggregated by a robust mean (discarding the highest and lowest facets before averaging) into 631 sample-level spectra; a second-generation spectrometer from the same manufacturer was used, 591–1001 nm with 229 channels.

Spectral outlier detection was applied to each season’s data: spectra were first standardised band by band, and then, using the Euclidean distance to the mean (i.e. the Mahalanobis distance under a diagonal

covariance assumption) at a fixed quantile, the most outlying 10% (63 samples in each season) were removed, leaving 567 samples for 2018 and 568 for 2025. The origin composition was: 2018, Xinjiang (arid plateau) 181, Shandong (temperate coastal plain) 186 and Shaanxi (semi-arid Loess Plateau) 200; 2025, Xinjiang 171, Shandong 177 and Gansu (semi-arid) 220. (After removal of samples with anomalous readings, Shaanxi in 2018 numbered 199 and Gansu in 2025 numbered 219.) On this basis, two further hard rules independent of sample size were applied to remove samples of doubtful quality; both are implemented uniformly at the data-reading entry point, and all experiments share the same data. First, samples with a reference SSC above 20°Brix were removed as anomalous readings (one in 2018, one in 2025): the physiological upper limit of SSC in mature ‘Fuji’ apples is about 20°Brix , whereas the measured values were 56.07 and 37.12°Brix , which can only be refractometer reading errors. Second, bit-wise identical spectra were removed (31 in 2018, none in 2025): the 2018 raw acquisition contained 18 groups of numerically identical spectra whose reference SSC values nevertheless differed (for example, three identical spectra corresponded to 12.13 , 14.17 and 14.37°Brix), all of which occurred at consecutive acquisition indices, whereas the 2025 data contained no such group—we judge this to be a spectrum-not-refreshed defect in acquisition, in which a later fruit inherited the spectrum of the preceding one. A single spectrum cannot simultaneously belong to fruits of different sugar content, so at least one spectrum–label pairing in each group is mismatched and there is no way to determine which; we therefore remove each group in its entirety rather than retaining any one member, so as to avoid making an unverifiable assumption about the direction of the defect.

After this processing, the final experimental samples were 535 for 2018 (Shandong 182, Xinjiang 158, Shaanxi 195) and 567 for 2025 (Shandong 177, Xinjiang 171, Gansu 219), 1102 in total. The six origin–year domains constitute all the experimental units of this study.

It should be noted that the 10% removal above is a percentile removal at a fixed proportion rather than a conditional removal based on an independent outlier criterion; it systematically inflates the absolute performance of all methods. Because all methods are compared on the same data, this operation does not affect the relative conclusions between methods—an expectation we confirm empirically in Section 3.5 by re-running the full benchmark at 0% and 15% removal, which leaves the method ranking unchanged.

2.5 Apple case study: spectral preprocessing

Spectra from the two seasons were aligned to 229 common wavelength channels within 591–1001 nm (selected by nearest wavelength). Wavelet denoising (db4 wavelet, 3 levels of decomposition), Savitzky–Golay smoothing (window 11, polynomial order 3) and band-wise standardisation were then applied in sequence. Scatter-correction preprocessing (multiplicative scatter correction, standard normal variate transformation and the like) was deliberately omitted from the main experiments so as not to erase in advance the very cross-origin scattering differences under study.

One physical qualification must be stated explicitly here: the spectra used in this study are standardised diffuse-reflectance spectra without an absorbance ($-\lg R$) transform, so their values are distributed on both sides of zero and are not absorbances in the strict sense. The Beer–Lambert factorisation below should therefore be understood as a non-negative low-rank additive structural prior *inspired by* the BL law[52], not as a strict physical decomposition of absorbance; correspondingly, the reconstruction loss acts on the magnitudes of the standardised spectra. The value of this prior is delimited by experimental evidence—cross-domain reconstruction residuals correlate significantly with transfer risk, and its low-rank additive factorisation architecture confers a limited accuracy advantage over a generic network in the uncorrected setting—rather than guaranteed by physical rigour. This qualification is also why we make no chromophore assignment for the latent components (see Section 3.4 and the Discussion).

Regarding the cross-domain protocol for standardisation: in cross-domain experiments the standardiser is fitted on the source domain only and then applied to the target domain, so that target-domain statistics do not enter the training pipeline (see Section 2.6).

2.6 Apple case study: design of cross-origin transfer scenarios and label budgets

For each ordered domain pair, four transfer types were defined—one source $\rightarrow$ one target, one source $\rightarrow$ multiple targets, multiple sources $\rightarrow$ one target, and multiple sources $\rightarrow$ multiple targets—giving 602 scenarios in total. The composition of these scenarios needs to be made explicit: only 24 of them are same-year cross-origin scenarios, while the remaining 578 span the 2018 and 2025 seasons (and hence two instrument generations). “Cross-origin transfer” in this paper is therefore a generalised domain shift that includes cross-year and cross-instrument components; Section 3.5 verifies the robustness of the core conclusions to scenario type by stratifying accordingly. The source domain is used for training in its entirety; the target domain is split 60%/20%/20% into training,

validation and test subsets. It should be noted that “20% target labels” is not a negligibly small sample in absolute terms: a single-origin target domain corresponds to roughly 30–40 calibration samples and a multiple-origin target domain to as many as a hundred or more, so the amount of target labelling on which the correction comparison of Section 3.7 depends is of the same order as the “small number of calibration samples” in the classical calibration-transfer literature.

Adaptation protocol: the source domain provides the supervisory signal; the target-domain training subset participates only in unlabelled form, through the domain-alignment and reconstruction losses, and does not enter regression supervision; the target-domain validation subset is used for early stopping and best-weight selection; and the test subset is used only for the final evaluation.

The label budget must be stated explicitly, because it directly affects the comparability of Table 6. Although the three deep models (CNN, CNN+MMD, Phys+BL) do not use target-domain labels as regression supervision, their early stopping and best-weight selection do use the labels of the target-domain validation subset, that is, they consume 20% of the target-domain label budget; PLSR and SVR, by contrast, are fitted on the full source domain and transferred directly, using no target-domain labels at all. The setting is therefore not a strictly zero-label transfer, and the label budget in Table 6 is asymmetric between methods—an asymmetry that favours the deep models. We stress that this asymmetry does not weaken the core conclusion but strengthens it: even though the deep models enjoy the advantage of 20% target-domain labels, that advantage disappears once PLSR is given the same budget to perform a single slope/bias correction (Section 3.7).

To render the deep models and the chemometric baselines comparable under the same label budget, the controlled experiment of Section 3.7 applies a slope/bias correction symmetrically to all methods, based on the same 20% of target-domain validation labels. In addition, to test whether the deep models’ use of 20% target labels for early stopping constitutes an unfair advantage, we add a strict zero-label control over all 602 cross-origin scenarios, in which the deep models instead use a source-domain hold-out set for early stopping and touch no target-domain labels at all (Section 3.5).

Because the same target set is reused under different source combinations (the 602 scenarios correspond to 62 unique target sets), statistical inference uses a cluster bootstrap; the scenario is the primary clustering unit (Section 2.9), and the more conservative unique-target-set and unique-source-set units are reported as robustness checks (Section 3.5), which together bound the spurious significance that reuse would otherwise produce. PLSR and support vector regression (SVR)

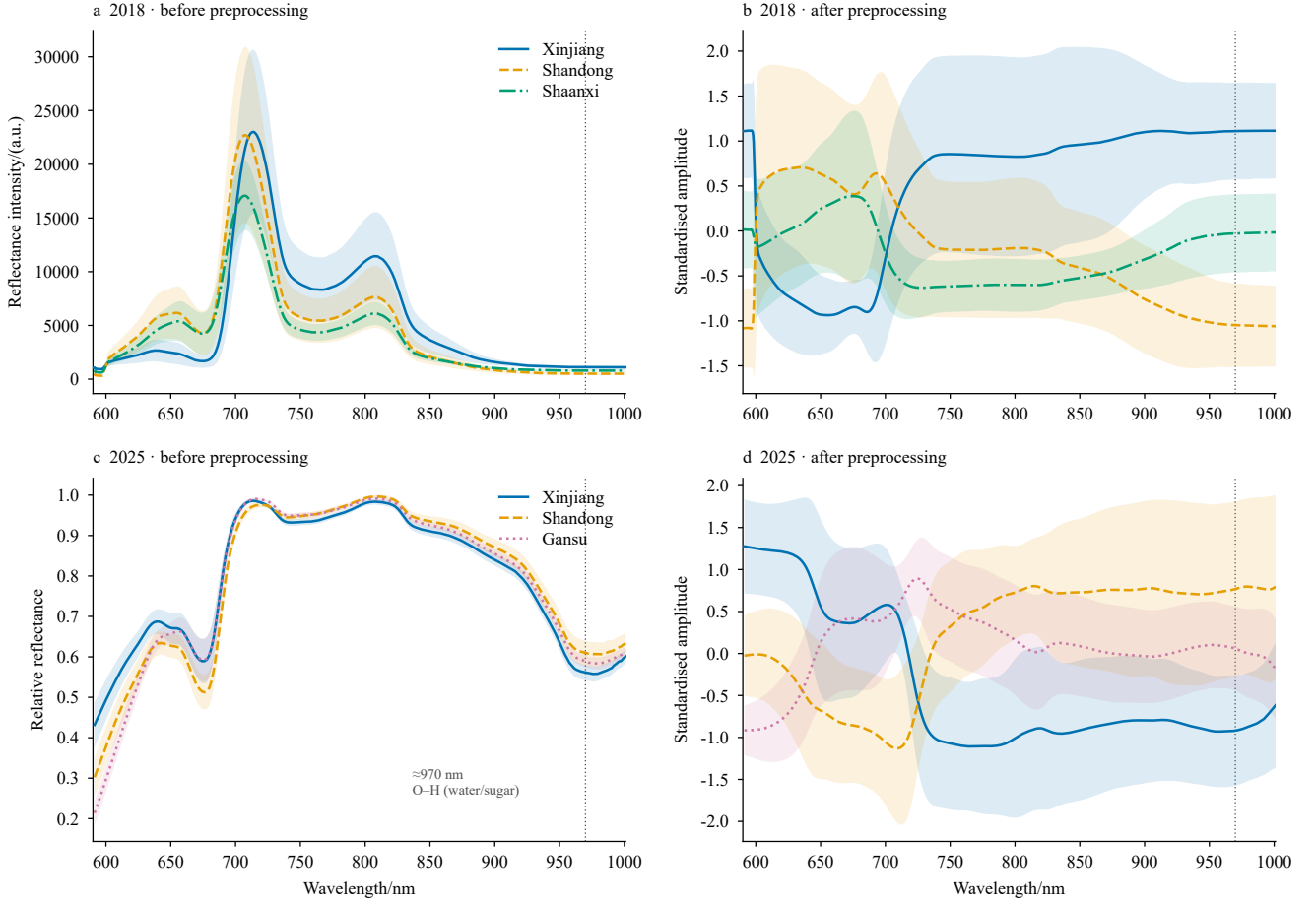


Figure 1: Apple NIR diffuse-reflectance spectra before and after preprocessing: 2018 (a, before; b, after) and 2025 (c, before; d, after); per-origin mean spectra with ± 1 SD shaded bands

Note: after preprocessing = wavelet denoising (db4, 3 levels) + Savitzky–Golay smoothing (window 11, order 3) + band-wise standardisation; scatter-correction preprocessing (MSC, SNV) is deliberately omitted (Section 2.5), so systematic structural differences between origins persist after preprocessing. The dotted vertical line marks the O–H overtone/combination band near 970 nm (associated with water and soluble sugars[53, 54]). The 2018 and 2025 data were acquired on different instruments with different raw signal units, so the two rows are scaled on separate ordinates.

cover all 602 scenarios; owing to a final batch of size 1 in batch normalisation and to occasional numerical divergence, the numbers of effectively converged scenarios for the deep models are 572 for the convolutional neural network (CNN), 568 for CNN+maximum mean discrepancy (MMD) and 568 for Phys+BL. Paired comparisons are restricted to the 568 scenarios in which both CNN and Phys+BL converged.

2.7 Apple case study: quantifying Beer–Lambert decomposability and the physics-informed factorisation architecture

We quantify BL-type additive decomposability by a low-rank surrogate. Let X_s be the source-domain spectral matrix; a K' -dimensional principal-component basis is fitted on it (mean-centring only, without band-wise scaling, so as to preserve the variance structure across bands), giving a reconstruction operator P . For any spectrum x , the relative reconstruction residual ΔR and the cross-domain amplification ratio A are defined as:

$$\Delta R(x) = \|x - P(x)\|_2 / \|x\|_2 \quad (1)$$

$$A = \text{median}\{\Delta R(x) : x \in X_t\} / \text{median}\{\Delta R(x) : x \in X_s\} \quad (2)$$

where X_t is the target-domain spectral matrix and K' is set to 10. $A > 1$ indicates that the target-domain spectra depart further from the low-rank additive structure of the source domain. We stress that computing A uses only spectra and no reference labels—this is the precondition for its use as a label-free indicator of transfer risk. Its robustness to the choice of low-rank basis, and whether it genuinely predicts cross-domain risk, are tested in Section 3.4.

The BL-inspired differentiable factorisation retains the “additive components $\times$ path length” skeleton while relaxing strict identifiability so as to absorb the wavelength-dependent non-linearity introduced by

scattering, reconstructing the input spectrum as a sum of products of K generalised component spectra, a scalar effective path length and generalised concentrations:

$$\hat{x}_i(\lambda) = \sum_{k=1}^K a_k(\lambda) b_i c_{ik}, \quad a_k, b_i, c_{ik} > 0 \quad (3)$$

where $a_k(\lambda)$ is the k -th generalised component spectrum, b_i the scalar effective path length of sample i and c_{ik} the corresponding generalised concentration; all three are constrained to be positive through an exponential transform, so as to preserve the form of additive mixing. K is set to 3, a pragmatic choice motivated by there being roughly three dominant sources of variation in intact-fruit visible–near-infrared spectra (absorption by water and by soluble sugars, and broadband scattering), without assigning any latent component to a specific constituent. We stress that this factorisation is not uniquely identifiable at the molecular level; moreover, because the input is not absorbance (Section 2.5), the learned components should all the less be interpreted as molar absorptivities or molar concentrations—they serve solely as a structural prior in latent space.

The network architecture is shown in Fig. 2: (1) a shared encoder (two fully connected layers, hidden dimension 64, batch normalisation + ReLU); (2) a three-head factoriser (component-spectrum head, path-length head and concentration head, made positive by an exponential transform); and (3) a two-layer multilayer perceptron regression head on the aggregated physical features, outputting SSC. The magnitude of the input spectrum is reconstructed according to the factorisation and used for the BL reconstruction loss.

The training objective is a weighted sum of the mean squared error of SSC regression and the reconstruction loss. The reconstruction loss is computed in log space to prevent the product terms of Eq. (3) from causing gradient explosion; because the input is a standardised spectrum (containing negative values), the absolute value is taken before the logarithm. We note that this $\ln(|x|)$ operation is a numerical device for stable optimisation of the multiplicative form in Eq. (3), not itself a physical relation: taking the absolute value of a mean-centred standardised spectrum discards sign information, so Eq. (4) should be read as a log-domain magnitude-reconstruction term rather than as a Beer–Lambert quantity. The Beer–Lambert motivation resides in the additive factorisation *architecture* of Eq. (3), not in this loss.

$$L_{\text{BL}} = \text{MSE}(\ln(\hat{x} + \varepsilon), \ln(|x| + \varepsilon)), \quad \varepsilon = 10^{-6} \quad (4)$$

$$L = L_{\text{reg}} + \lambda_{\text{BL}} [L_{\text{BL}}(X_s) + L_{\text{BL}}(X_t)] \quad (5)$$

Here the regression loss term is the mean squared error of source-domain SSC prediction; the BL reconstruction loss is summed symmetrically over the source and target domains (the target domain participating in unlabelled form only, see Section 2.6); and the default BL loss weight is 0.5. The optimiser is Adam (learning rate 1×10^{-3} , batch size 16, at most 200 epochs), with early stopping on the RMSE of the target-domain validation subset (patience 20).

2.8 Apple case study: comparison methods

PLSR uses 10 latent variables, fitted on the full source domain and transferred directly (no adaptation). SVR uses a radial basis function kernel ($C=100$, $\varepsilon=0.1$) and is likewise transferred directly. The CNN consists of two one-dimensional convolutional blocks (kernel widths 5 and 3, channels $64 \rightarrow 32$), max pooling, dropout (0.3), two encoding layers and a two-layer regression head, trained with source-domain supervision. CNN+MMD adds to the CNN a maximum mean discrepancy term between source and target latent features[30], with weight 0.1. All methods use the same source/target split and random seed within each scenario, ensuring fair pairing.

2.9 Apple case study: evaluation metrics and statistical tests

We report the root mean square error (RMSE), the coefficient of determination (R^2), the ratio of performance to deviation (RPD) and the bias. The quality misclassification rate is defined as the proportion of test samples whose predicted binary grade (acceptable, $\geq 13^\circ\text{Brix}$; unacceptable, $< 13^\circ\text{Brix}$) disagrees with the reference grade.

Randomness and clustering: every scenario in Tables 6 and 8 was repeated independently under five fixed random seeds, the seed controlling both the target-domain data split and the network initialisation. The multiple seeds of a given scenario are repeated measurements rather than independent samples, so the across-seed mean is taken first for each scenario and the scenario is then used as the statistical unit—that is, the clustering unit is the scenario and the sample size is the number of scenarios, not “scenarios $\times$ seeds”. The across-seed standard deviation is reported separately to reflect the sensitivity of the models to randomness.

Paired comparisons use the two-sided Wilcoxon signed-rank test (as the primary test, the RMSE distribution being heavy-tailed and non-normal), with the paired t -test as an auxiliary; multiple comparisons are corrected by the Holm procedure within a pre-specified family of hypotheses; effect sizes are reported as Cliff’s δ (non-parametric) and Cohen’s d ; and confidence intervals come from a cluster bootstrap with the scenario as the clustering unit (1000 resamples), not an i.i.d. bootstrap. Given the heavy-tailed distribution of errors across scenarios, the coefficient of determination (R^2) is

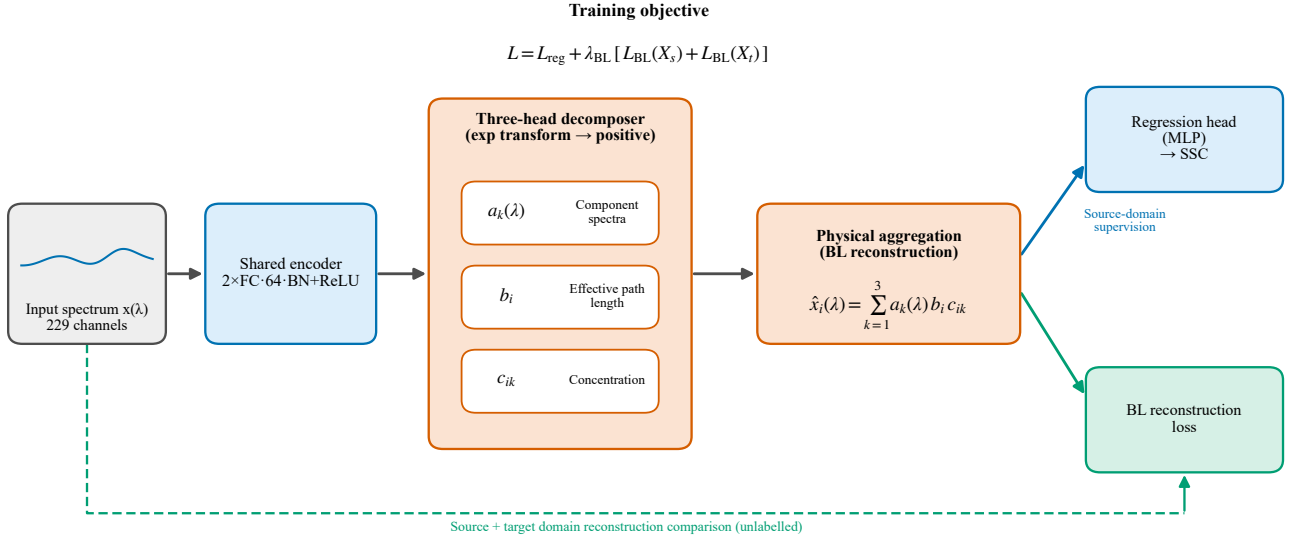


Figure 2: Forward flow and training objective of the physics-informed differentiable factorisation architecture (Phys+BL)

reported in the tables as the across-scenario median—its mean being severely distorted by a few divergent scenarios—while RMSE, RPD and bias are reported as means.

2.10 Apple case study: a calibration-transfer control experiment under a unified data split

When a small number of calibration labels is available in the target domain, classical calibration-transfer methods are already competitive. To delimit the scope of application of our method under fully comparable conditions, we add the following control experiment. (1) Unified data split: all methods are evaluated under the same random split (the same random number generator and seed, reset for each scenario), eliminating any incomparability due to inconsistent splits. (2) Post-hoc slope/bias correction: this correction is a classical means of calibration transfer and is applicable to the predictions of any model. On the target-domain validation subset (20% of the target domain, i.e. the same set of labels used for early stopping by each deep model), a univariate linear regression of reference values on predictions is fitted by least squares, and the resulting slope a and intercept b are used to correct the predictions on the test subset:

$$(a, b) = \arg \min_{a, b} \sum_{i \in \text{val}} (y_i - a\hat{y}_i - b)^2, \quad \hat{y}' = a\hat{y} + b \quad (6)$$

where $\hat{y}$ is the model's raw prediction on a test sample and $\hat{y}'$ the corrected prediction. If the standard deviation of the predictions on the validation subset is close to zero (a degenerate case), the correction is abandoned and the raw predictions are used. (3)

Symmetric application: the correction is applied equally to PLSR, SVR, CNN and Phys+BL, forming a 2×4 uncorrected/corrected contrast, in order to answer two key questions—whether the physical prior retains an advantage over classical calibration transfer when a small number of target-domain labels is available, and whether the advantage of the physical architecture over a generic CNN survives the correction.

3 Results

3.1 Label-budget reckoning between simple corrections and deep/physics representations

Under the unified protocol, the three families of methods were compared on the 126 transfer tasks of the five calibration-transfer benchmarks in terms of target-domain root mean square error at each target-label budget (median across the five fixed seeds, with the task as the clustering unit). A first, descriptive comparison takes the task-wise paired difference between the *best simple method* (label-free second-order moment alignment CORAL or the two-parameter slope/bias correction SBC) and the *best deep/physics method* (one-dimensional convolutional network, domain-adversarial DANN, DeepCORAL, or the physics-informed Beer–Lambert factorisation Phys+BL), tested by the Wilcoxon signed-rank test with effect size reported as Cliff's δ (Table 4). Because this takes the per-task best within each family, it is an *oracle upper envelope* for both families rather than an operational selection rule, and the families are of unequal size (two simple, four deep), so the envelope

should be read as descriptive; a pre-specified, selection-free reanalysis that confirms the pattern is reported below (Table 5).

The results show distinct patterns according to the type of domain shift. For **instrument-response shift** (corn across three instruments and tablets across two, 30 tasks), the simple corrections are consistently better than the deep representations at every label budget: median root mean square errors of 0.24–0.30 against about 0.50 for the deep methods, with the simple methods better on 90%–97% of tasks, δ between 0.46 and 0.48 (a large effect) and Wilcoxon $P < 10^{-4}$. For **seasonal shift** (mango across four seasons, 12 tasks), the simple methods are likewise better at most budgets (δ 0.31–0.67, $P \leq 0.02$). For **cross-origin/year shift** (apple, 72 tasks), the envelope comparison is mixed: at zero to five target labels the best-deep envelope is marginally lower (median difference -0.05 to -0.13°Brix , $P = 0.01/0.001$), but this margin disappears at 10 labels and, at 20–40 labels, the simple methods are the better of the two (δ 0.17–0.18, $P < 10^{-3}$). As shown below, this near-zero-label deep margin does not survive a selection-free comparison and appears attributable to the larger deep family. For **cross-laboratory mid-infrared soil** (12 tasks), the two families are not statistically distinguishable ($P = 0.30$ –0.62); with only 12 tasks this is an absence of evidence of a difference rather than established equivalence.

In summary, on these benchmarks the more expensive deep and physics-informed representations were not found to be better than the simplest classical corrections under cross-instrument and cross-season transfer; under cross-origin transfer the best-deep envelope held at most a marginal edge at near-zero labels and was overtaken as the budget grew; and on cross-laboratory soil the two families were not distinguishable. The physics-informed Beer–Lambert architecture (Phys+BL), as a member of the deep group, did not consistently beat the simple corrections under any type of shift, consistent with the in-depth observations on the single apple benchmark (case study in Section 3.4). The complete reckoning curves of the three families against label budget under the four types of shift are shown in Fig. 3.

To check that these patterns are not an artefact of taking the per-task best within each family (a winner’s curse, compounded by the four-versus-two family-size imbalance), we recomputed the comparison on the same runs in two selection-free ways (Table 5): (i) a *pre-specified representative pair* at zero labels, CORAL against its deep counterpart DeepCORAL, both being second-order moment alignment and therefore the closest simple-versus-deep analogue; and (ii) the *family mean*, averaging over all methods in each family per task rather than taking the best. Neither uses post-hoc selection. Under both, the pattern is preserved and,

if anything, slightly clearer: simple methods retain a large paired advantage under instrument shift (family-mean Cliff’s δ 0.93 at both 0 and 20 labels) and a clear one under seasonal shift (δ 0.50–1.00). Most notably, the marginal near-zero-label deep edge seen in the envelope under cross-origin shift ($\delta = -0.09$) does not persist: both the representative pair ($\delta = +0.28$, simple better on 64% of tasks, $P = 0.003$) and the family mean ($\delta = +0.25$, 62%, $P = 0.02$) instead favour the simple methods there, indicating that the envelope’s apparent deep advantage at near-zero labels was largely a selection effect over the larger deep family. On cross-laboratory soil the two families remain statistically indistinguishable under both analyses ($\delta = +0.17$, $P > 0.4$). The selection-free comparison thus supports the same qualitative reading as the envelope while removing its most obvious bias. Nor is the choice of DeepCORAL as the comparator responsible: on the same zero-label runs, CORAL individually outperforms *each* of the four deep methods (CNN, DANN, DeepCORAL and Phys+BL)—on 97–100% of tasks under instrument shift (Cliff’s δ 0.48–0.49, all $P < 10^{-8}$) and on 68–75% of tasks pooled across the 126 tasks (all $P < 10^{-5}$)—so the representative pair is representative rather than selected.

Two further robustness checks address the incommensurate-units and pseudo-replication concerns of pooling across benchmarks (script 68). First, the family comparison was recomputed on the dimensionless normalised error $\text{NRMSEP} = \text{RMSEP}/\sigma_{y,\text{tgt}} = 1/\text{RPD}$, which is comparable across analytes and datasets; the pattern is unchanged (family-mean Cliff’s δ 0.93 for instrument, 0.50–1.00 for season, +0.25 for origin/year at zero labels, and not distinguishable for soil). Second, taking the benchmark as the unit of analysis—the most conservative choice, giving only $n = 5$ independent units—all five benchmarks favour the simple family at both 0 and 20 target labels; a benchmark-balanced bootstrap gives a mean NRMSEP difference (deep minus simple) of +0.21 (95% CI [0.06, 0.38]) at zero labels and +0.23 ([0.11, 0.38]) at 20 labels, both excluding zero, although two benchmarks (apple and soil) are near-ties at the benchmark level. The reading is therefore not an artefact of the error unit or of within-benchmark replication, though with $n = 5$ benchmarks the benchmark-level test is necessarily low-powered.

3.2 The role of label-free alignment and an in-sample “shift type $\times$ label budget” frequency map

We next separate zero-label feasibility from the budget dependence of the best method. **Label-free second-order moment alignment (CORAL)** recovers accuracy substantially without using any target labels, and the extent of the recovery depends on the type of shift: for instrument-response shift, zero-label CORAL raises the median ratio of performance to

Table 4

Oracle upper-envelope comparison: best simple correction vs. best deep/physics representation, paired by task (median across the five fixed seeds, Cliff's δ and the paired Wilcoxon test). Because the per-task best is taken within each family, this is a descriptive upper envelope, not an operational selection rule; a selection-free reanalysis is given in Table 5.

Shift type	Label budget	Tasks	Simple median	Deep median	Simple-win %	Cliff's δ	Wilcoxon P
Instrument	0	30	0.241	0.500	97	0.48	$< 10^{-4}$
Instrument	20	30	0.283	0.512	90	0.48	$< 10^{-4}$
Instrument	40	30	0.279	0.504	97	0.47	$< 10^{-4}$
Season	10	12	2.037	2.561	92	0.60	0.002
Season	20	12	1.975	2.475	100	0.60	0.001
Origin/year	0	72	1.966	1.740	39	-0.09	0.010
Origin/year	5	72	1.457	1.468	28	-0.09	0.001
Origin/year	20	72	1.285	1.326	64	0.18	$< 10^{-3}$
Laboratory (MIR)	10	12	11.270	11.958	33	0.00	0.301

Note: units are those of the target variable of each benchmark ($^{\circ}$ Brix or dry matter % for apple/mango, constituent % for corn, total carbon % for soil). "Simple" is the better of CORAL and SBC on each task; "deep" is the best of CNN/DANN/DeepCORAL/Phys+BL on each task. "Simple-win %" is the percentage of the paired tasks on which the simple family has the smaller error—the paired complement to Cliff's δ ; a negative δ (win below 50%) indicates that the deep methods are better. All results use the five fixed seeds [20060515, 20041210, 19810915, 2023, 2024].

Table 5

Selection-free comparison of the simple and deep families (no per-task best-of-family selection), computed from the same runs as Table 4. Two variants: a pre-specified representative pair (label-free CORAL vs. its deep counterpart DeepCORAL) and the family mean (mean over all methods in each family per task). A positive Cliff's δ and a win fraction above 0.5 both indicate that the simple family has the smaller error.

Comparison	Shift type	Budget	Simple med.	Deep med.	Simple-win %	Cliff's δ	Wilcoxon P
CORAL vs. DeepCORAL	Instrument	0	0.241	0.500	100	+1.00	$< 10^{-8}$
CORAL vs. DeepCORAL	Season	0	1.959	2.616	75	+0.50	0.016
CORAL vs. DeepCORAL	Origin/year	0	1.966	1.926	64	+0.28	0.003
CORAL vs. DeepCORAL	Laboratory (MIR)	0	11.33	15.18	58	+0.17	0.52
Family mean	Instrument	0	0.241	0.502	97	+0.93	$< 10^{-8}$
Family mean	Instrument	20	0.283	0.519	97	+0.93	$< 10^{-5}$
Family mean	Season	20	1.975	2.509	100	+1.00	$< 10^{-3}$
Family mean	Origin/year	0	1.966	1.905	62	+0.25	0.018
Family mean	Origin/year	20	1.285	1.529	82	+0.64	$< 10^{-9}$
Family mean	Laboratory (MIR)	20	11.08	13.22	58	+0.17	0.47

Note: same units and seeds as Table 4. "Simple-win %" is the percentage of paired tasks on which the simple family has the smaller error (the paired complement to Cliff's δ). The representative pair is defined a priori as the two second-order moment alignment methods (CORAL, DeepCORAL) and is compared at zero labels, where both are label-free. The family mean is the per-task *unweighted* mean over each family's constituent methods (simple: CORAL, SBC; deep: CNN, DANN, DeepCORAL, Phys+BL with their zero-shot and fine-tuned variants), so the deep family is summarised over more constituents than the simple family. Only budgets 0 and 20 are shown; the full grid (script 67) gives the same direction at 5/10/40—family-mean δ is +0.36 to +0.61 for origin/year at 10–40 labels and remains large positive for instrument and season, the only near-tie being origin/year at 5 labels ($\delta = -0.08$, $P = 0.94$).

deviation (RPD) from 0.61 for direct transfer of the source model to **2.01**, reaching a common screening-level threshold ($\text{RPD} \geq 2$ is an empirical, application- and variance-dependent screening convention rather than a guarantee of industrial-grade accuracy[51], and 2.01 is a benchmark-level median: about 53% of the 30 instrument tasks individually reach $\text{RPD} \geq 2$ and 83% reach $\text{RPD} \geq 1.4$); for seasonal, cross-laboratory and cross-origin shift, CORAL raises RPD from 0.84, 0.72 and 0.49 to 1.27, 1.02 and 0.74 respectively, a partial recovery that falls short of that threshold. It should be noted that CORAL is a transductive adaptation: it estimates its second-order moments from the unlabelled target spectra, so "zero labels" means no target *labels*,

not no target *data*, and the alignment is refitted per target domain rather than deployed inductively to a single new sample.

Organising the *most frequent winning method* at each label budget under each type of shift into an in-sample frequency map (Fig. 4): at zero labels, label-free alignment (CORAL) is the most frequent winner under every type of shift; as target labels accumulate, instrument shift moves to model updating and population/seasonal shift to the two-parameter slope/bias correction, until, with sufficient labels, a target-only model (target PLS alone) catches up. Read as a practical heuristic rather than a validated rule, this suggests first determining the type of shift and the available label

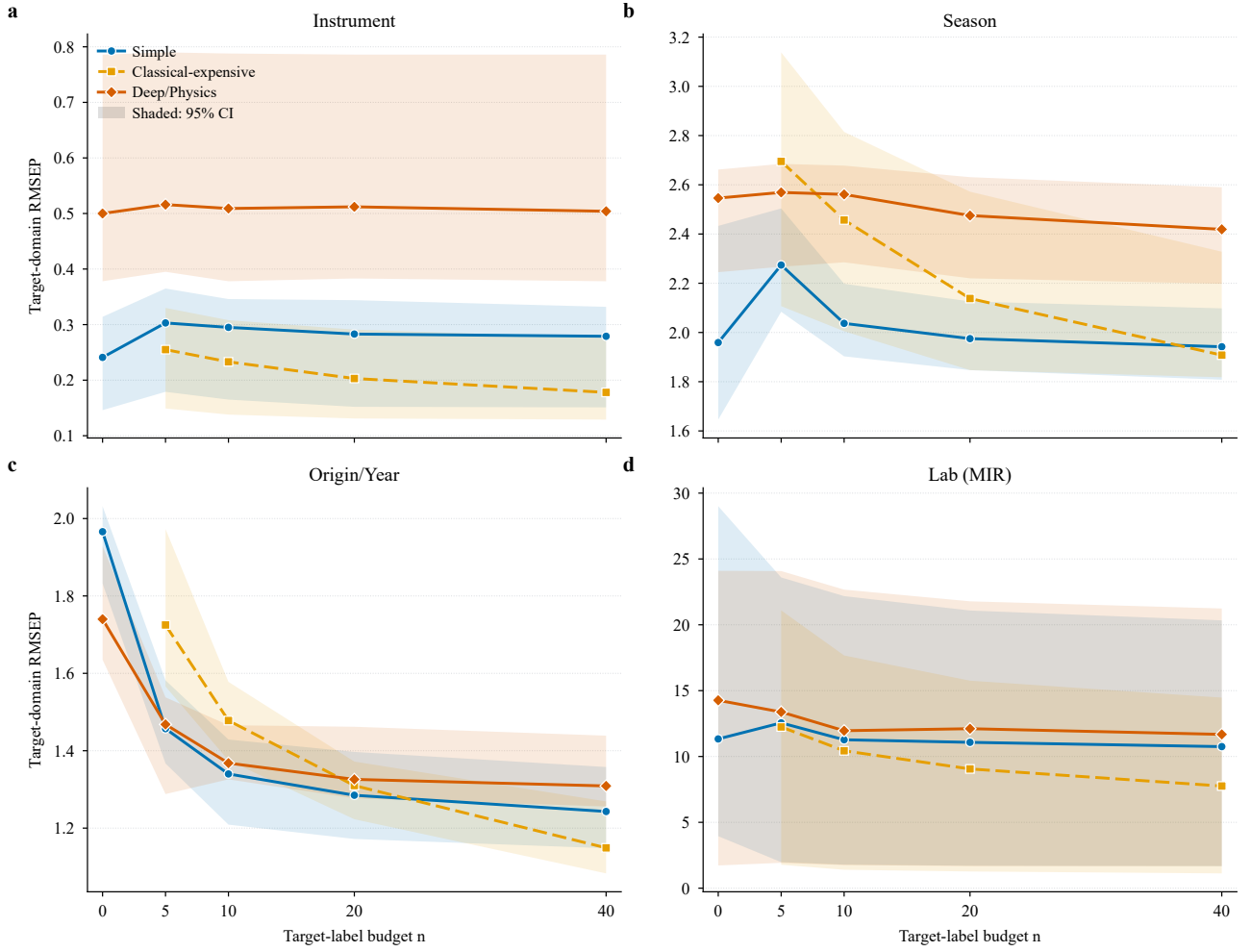


Figure 3: Reckoning curves of the three method families (best simple correction, classical model update, best deep/physics representation) against target-label budget under the four types of domain shift: median target-domain RMSEP with task-clustered 95% confidence bands

Note: the ordinate of each panel is in the native target units of the corresponding benchmark (instrument = corn constituents in % and tablet weight/hardness/assay in their own units; season = mango dry matter %; cross-origin/year = apple °Brix; cross-laboratory = soil total carbon and sand in %); the units differ, so the axes are scaled separately and values are not comparable across panels. Medians are taken across the five fixed seeds, and the shaded bands are cluster-bootstrap 95% confidence intervals with the task as the clustering unit.

budget, then trying the simplest corresponding method from the table before deploying a deep model. Because the table is fitted on these same five benchmarks, it requires prospective validation before being relied on (see Section 4).

3.3 The crossover budget for “when simple is enough” cannot be predicted from measures of domain shift

The patterns above naturally raise a question: can the crossover label budget n^* at which a simple method catches up with or overtakes a complex one be predicted in advance from measurable quantities of domain shift, thereby forming a “transferability scaling law”? We measured n^* for each transfer task and regressed it in log-log space on the magnitude of domain shift (maximum mean discrepancy MMD, sliced Wasserstein

distance), the intrinsic dimensionality of the target domain and the signal-to-noise ratio; extrapolability was then tested by leave-one-benchmark-out extrapolation (fit on the remaining benchmarks, predict the held-out benchmark).

The answer is *negative*: the log-log fits of n^* on the above continuous measures of shift have coefficients of determination R^2 of only 0.002–0.13 (at most 0.25 under the n_{usable} definition, and then on only 27 tasks for which it is defined), leave-one-benchmark-out extrapolation reaches a mean absolute error of 3.6–34 labels on the original scale, and the coverage of conformal 90% intervals is erratic, between 0.27 and 1.00. The scatter of n^* against MMD (Fig. 5) is nearly flat, illustrating this negative result directly. Even with the cross-modality mid-infrared soil benchmark included and five fixed seeds used throughout,

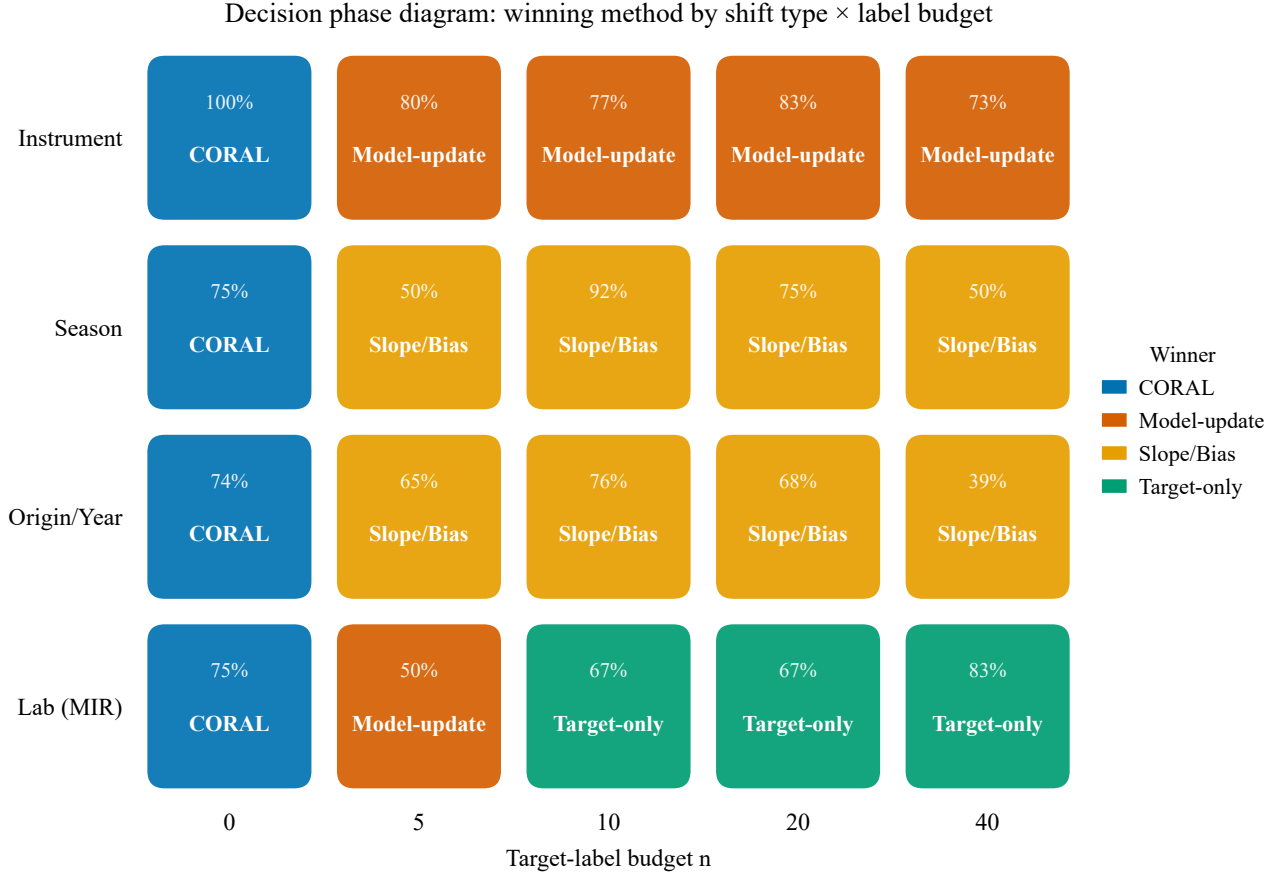


Figure 4: In-sample frequency map over shift type $\times$ label budget: each cell is annotated with the most frequent winning method on this study’s data and its share among tasks of that type; this is a retrospective in-sample summary, not a validated decision rule

Note: the winning method is the one attaining the smallest target-domain RMSEP on the largest number of tasks under that (shift type, label budget) combination; at zero labels the most frequent winner under every shift is label-free second-order moment alignment (CORAL), which is transductive and requires access to the unlabelled target spectra. This is an in-sample frequency map derived from five benchmarks (cell counts as small as 12 for season/lab), and the “winner” inherits the same best-of-family selection caveat as Table 4; it should be read alongside the selection-free comparison of Table 5 and validated prospectively before use, not as a universal rule.

this conclusion—that *no* extrapolable scaling relation is found—remains robust. A possible explanation is that domain distances across different types of shift (instrument response, population, season, laboratory) are *nearly incommensurable* under the metrics used here—the same numerical MMD corresponds to widely different transfer difficulties under different mechanisms, so marginal distribution distances alone can hardly characterise n^* across mechanisms. The scope and strength of this negative result must be delimited carefully. First, it is confined to the family of metrics examined here (MMD, sliced Wasserstein, intrinsic dimensionality) and to the log–log functional form, and does not exclude the existence of a scaling relation under other metrics or representations (for instance in a self-supervised pretrained latent space). Second, the number of independent benchmarks is only $n = 5$ and each type of shift is carried mainly by a single dataset (see Section 3.1 and the collinearity limitation in the Discussion), while n^* is itself a discrete and

possibly censored quantity on the budget grid; this conclusion is therefore the empirical statement that *no unified scaling law was observed under the present design*, not a proof that none exists. What can be stably reproduced is only the ranking of methods over *discrete* types of shift (the look-up table of Section 3.2), not a quantitative extrapolation from continuous shift magnitude.

3.4 Apple case study: destabilisation of cross-origin Beer–Lambert additive decomposability

For 12 within-year one-source $\rightarrow$ one-target scenarios (six single-origin pairs in each of 2018 and 2025; the pairing is restricted to within a year so as to isolate origin differences and exclude cross-year instrument and season effects), a PCA-10 basis fitted on the source domain was used as the low-rank surrogate for BL decomposability, and the relative reconstruction

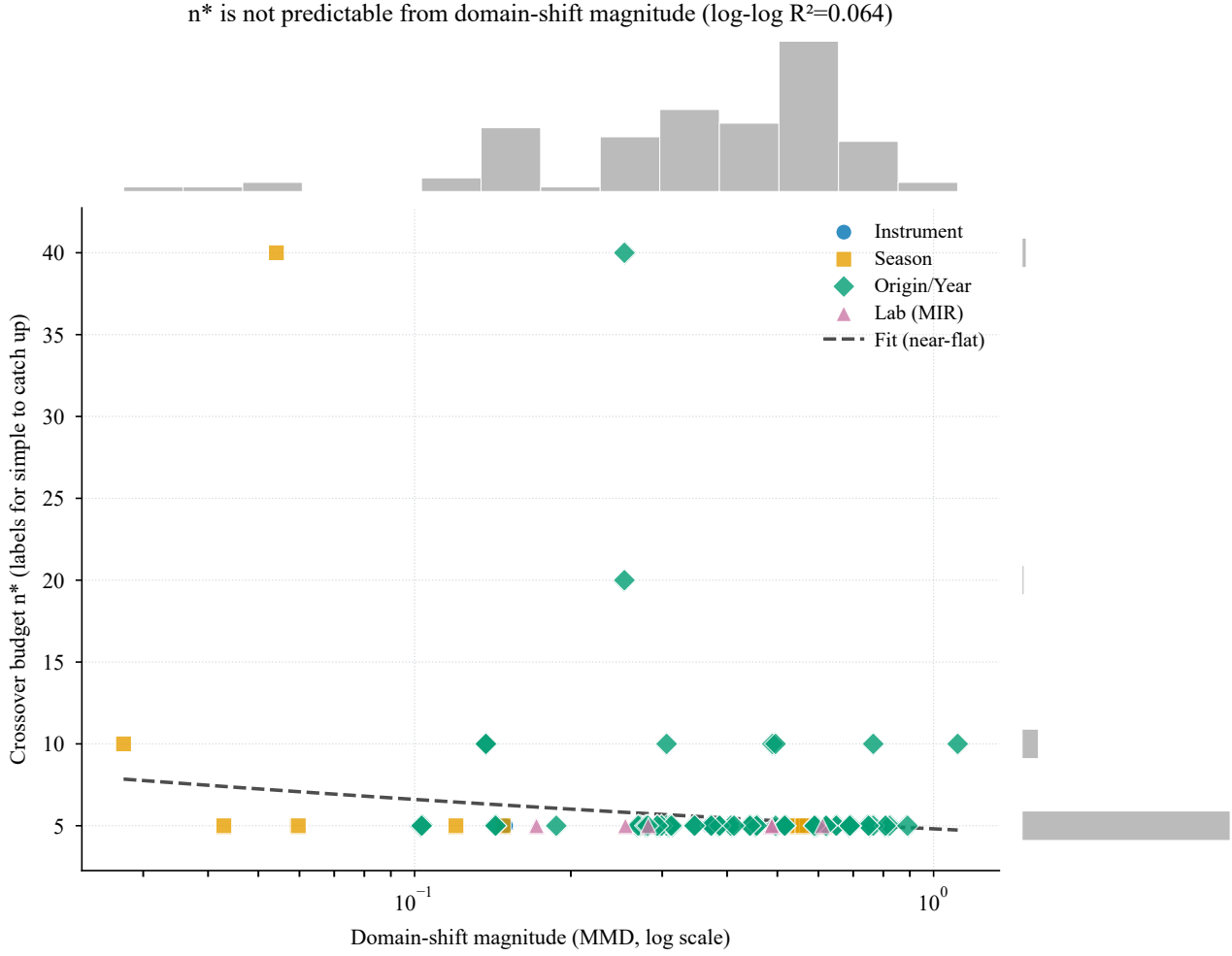


Figure 5: Crossover budget n^* (the number of target labels needed for the simple method to catch up with the complex one) against domain-shift magnitude (maximum mean discrepancy, MMD): the log-log fit is near-flat ($R^2 \approx 0.06$), coloured by shift type

Note: the abscissa is MMD on a logarithmic scale; the types of shift (instrument/season/origin-year/laboratory) are distinguished by colour; the dashed line is the log-log least-squares fit. The same numerical MMD corresponds to widely different n^* under different mechanisms, that is, domain shift is incommensurable across mechanisms (see the main text).

residuals were computed on the source domain (within-domain) and the target domain (cross-domain). All 12 scenarios show a systematic amplification of the residual, with ratios from 1.08 (Shaanxi→Shandong, geographically closest) to 4.55 (Xinjiang→Shaanxi); these 12 scenarios form the pre-specified family of tests, and all remain significant after Holm correction of the Mann–Whitney U test ($P < 0.05$; largest corrected $P = 0.021$) (Fig. 6a). Grouped by year, the cross-domain amplification trend holds independently in both acquisition years: the median target-domain residual is significantly higher than that of the corresponding source domain in each year (2018, 0.012→0.034, a 2.8-fold amplification; 2025, 0.059→0.100, a 1.7-fold amplification). The residual distributions of the individual scenarios are shown in Fig. 6a.

To exclude an artefact of within-domain overfitting, each source domain was split 70%/30% into a fitting

subset and a hold-out subset and re-estimated on the held-out source samples: 28 of 36 comparisons remain significant ($P < 0.05$), and the hold-out–target amplification ratios agree with the fitting–target ones. The pairwise Pearson correlations between the amplification ratios of three low-rank surrogates (PCA-10, NMF-3, SVD-3) are all at least 0.77 ($P < 0.005$) (Fig. 6b), indicating that this is a data-driven and robust phenomenon rather than a by-product of basis selection. Note that an earlier implementation included a further surrogate denoted NNLS-3, which shares the same non-negative basis as NMF-3 and differs only in the solver for the mixing weights; the two give bit-wise identical results (Pearson $r=1.000$) and NNLS-3 is therefore not counted as an independent surrogate.

We make no chromophore assignment for these low-rank components. The basis serves only as a low-rank surrogate for BL additive decomposability; the peak

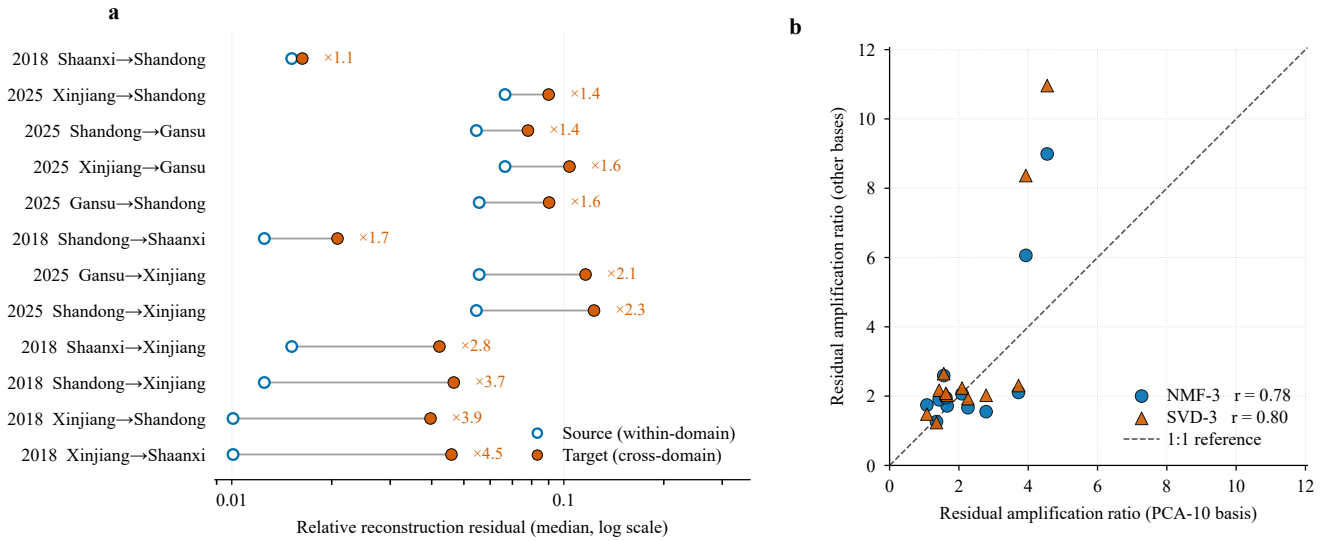


Figure 6: Systematic amplification of cross-origin Beer–Lambert reconstruction residuals (a) and consistency of the amplification ratios across low-rank surrogates (b)

Note: in (a), each row is one source→target cross-origin scenario (12 in total, sorted in ascending order of amplification ratio); open circles denote the median relative reconstruction residual on source samples and filled circles that on target samples, the connecting line indicates the direction of amplification, the value on the right is the amplification ratio, and the abscissa is on a logarithmic scale. In (b), the abscissa is the amplification ratio from the PCA-10 basis and the ordinate those from the NMF-3 and SVD-3 bases, the dashed line is the 1:1 reference and r is the Pearson correlation coefficient.

positions of its components vary with the data subset and the decomposition hyperparameters, and they have no stable chemical correspondence.

We further tested whether BL residual amplification can serve as a label-free predictor of transfer risk, by computing Spearman correlations between the scenario-level BL amplification ratio and the cross-domain root mean square error of each method. The correlations are significant and consistent in direction (median ρ across methods 0.43, range 0.22–0.66, all $P < 0.001$, with all 95% confidence interval lower bounds above zero), being strongest for partial least squares regression and the convolutional neural network ($\rho \approx 0.65$); under leave-one-origin-out cross-validation, the median ρ of the risk ranking remains positive for origins never used in fitting (0.17–0.68). It must be stated clearly, however, that this correlation strength does not reach our pre-registered diagnostic threshold ($\rho \geq 0.5$); the BL residual is therefore positioned as a label-free risk warning that correlates significantly with transfer risk, not as a diagnostic signal that can stand on its own—and this boundary is itself part of the “scope of application” theme of this paper.

3.5 Apple case study: cross-origin transfer performance of five methods

Table 6 reports the aggregate performance of the five methods on the cross-origin transfer scenarios. Read in absolute terms, every method fails on this benchmark: the across-scenario median R^2 is negative and the RPD below 1 for all five, so none reaches screening-level accuracy and the comparisons

that follow are relative rankings within a stratum of shared failure. Within that stratum, Phys+BL attains the lowest mean RMSE, 1.873 °Brix, 11.3% lower in aggregate than the CNN (2.109 °Brix); over the 568 common scenarios, the mean paired improvement is 9.88% and the median 13.42%, with an extremely small nominal P value from the two-sided Wilcoxon test ($P = 1.3 \times 10^{-67}$, after Holm correction). It must be noted, however, that the 602 scenarios are formed by reusing combinations of only six atomic origin–year domains (several origins in each of the 2018 and 2025 seasons) and are not 602 mutually independent experiments; nominal P values of this kind are systematically inflated by strong correlation between scenarios, so we take cluster-robust confidence intervals, rather than nominal P values, as the primary basis for inference. A cluster bootstrap with the scenario as the clustering unit gives a paired RMSE difference of -0.238 °Brix with a 95% confidence interval of $[-0.33, -0.12]$ °Brix (Cliff’s $\delta = -0.57$, a large effect).

This conclusion is robust to the choice of clustering unit. Besides the scenario (602), the cluster bootstrap was repeated with two more conservative clustering units: the target set (62 unique target-domain sets) and the source set (62 unique source-domain sets). For the paired difference of uncorrected Phys+BL relative to the CNN, the 95% confidence intervals under scenario/target-set/source-set clustering are $[-0.33, -0.12]$, $[-0.34, -0.11]$ and $[-0.39, -0.08]$ °Brix respectively; after correction, the differences of PLSR+SB and CNN+SB relative to Phys+BL+SB are likewise all negative under all three clustering units, with confidence intervals that never span zero (for example,

PLSR+SB relative to Phys+BL+SB gives $[-0.45, -0.12]$, $[-0.45, -0.13]$ and $[-0.45, -0.10]$ °Brix respectively). Even under the most conservative source-set clustering, neither the direction nor the significance of “uncorrected Phys+BL is best; after correction the classical methods overtake it” changes (the three-level clustering robustness analysis is in supplementary script 60). Stratifying into same-year cross-origin (24 scenarios) and cross-year including cross-instrument (578 scenarios), the core conclusions agree in both strata—Phys+BL is best when uncorrected, and PLSR+SB overtakes Phys+BL+SB after correction—with only the absolute difficulty being higher in the cross-year stratum (mean PLSR RMSE rising from 2.22 to 3.37 °Brix). The uncorrected difficulty structure of the one-source→one-target origin pairs is shown in Fig. 8; it is determined mainly by the target domain (transfer to 2025 Gansu being hardest and to 2018 Shandong easiest).

This ranking is also insensitive to the outlier-removal threshold. The 10% percentile removal of Section 2.4 is a fixed proportion chosen without reference to an independent outlier criterion, so we re-ran the full multi-seed benchmark (seven methods, five fixed seeds, 602 scenarios) at 0% (no removal) and at 15% removal in addition to the 10% used above. Because a handful of divergent scenarios severely distort the mean (Section 2.9)—at 0% removal the uncorrected Phys+BL mean RMSE inflates to 34.2 °Brix even though its median stays at 1.74 °Brix—threshold-robustness is assessed on the median. On that robust statistic the two core orderings are unchanged across all three thresholds: uncorrected, Phys+BL retains the lowest median RMSE (1.74, 1.71 and 1.71 °Brix at 0%, 10% and 15% removal respectively); after slope/bias correction, PLSR+SB falls below Phys+BL+SB (1.55/1.59/1.55 versus 1.62/1.62/1.62 °Brix), so the classical correction again overtakes the physics architecture. Only the absolute error level shifts with the removal fraction; the relative ranking—and hence every conclusion of this section—does not, which directly answers whether the fixed 10% choice drives the result. The inflation of the physics model’s mean under no removal is itself a direct manifestation of the residual divergence risk discussed in Sections 3.6 and 4.

To address concerns about classical chemometric baselines, we added direct transfer (zero target labels) with four common NIR preprocessing methods combined with PLSR (standard normal variate transformation SNV, multiplicative scatter correction MSC, and Savitzky–Golay first and second derivatives). The best of these, PLSR+SNV, gives a median RMSE of 2.01 °Brix, comparable to the generic convolutional network (median 2.02) but short of the 1.74 °Brix of the physics factorisation architecture when uncorrected; MSC and the second derivative are numerically unstable in some scenarios. After slope/bias correction,

PLSR+SB (1.59 °Brix) remains the best overall. Introducing classical preprocessing baselines therefore does not change our conclusions.

To address further the concern that the deep models’ consumption of 20% target labels for early stopping constitutes an unfair advantage, we added a strict zero-label control over all 602 cross-origin scenarios: all three deep models (CNN, CNN+MMD, Phys+BL) instead use a source-domain hold-out set for early stopping and touch no target-domain labels whatsoever, and are compared alongside the four classical preprocessing baselines, which likewise use zero target labels. Even stripped of target-label early stopping, Phys+BL still attains a median RMSE of 1.75 °Brix, the best overall, ahead of the best classical baseline PLSR+SNV (2.01 °Brix), the generic convolutional network (2.27 °Brix) and CNN+MMD (2.28 °Brix); under the strict zero-label protocol the numbers of effectively converged scenarios for CNN, CNN+MMD and Phys+BL are 566, 563 and 563 respectively (paired comparisons being restricted to the 563 scenarios in which all three converged). It is particularly notable that the median RMSE of Phys+BL under the strict zero-label protocol (1.75 °Brix) is almost identical to its level under 20% label early stopping (1.74 °Brix), indicating that the uncorrected advantage of the physics factorisation architecture is not due to target-label early stopping—consistent with the earlier statement that its advantage originates in the low-rank additive factorisation architecture itself. That advantage nonetheless remains confined to the uncorrected setting, and it is likewise a relative optimum within a set of methods that all fail.

The RPD of Phys+BL rises from 0.772 for the CNN to 0.913, the highest overall; its bias is -0.15 °Brix, whereas the systematic underestimation of SVR, the CNN and CNN+MMD is more pronounced (-0.60 to -0.85 °Brix).

It must be stressed, however, that this “best” is the best within a set of methods that all fail. The median coefficients of determination in Table 6 are all negative, meaning that in most scenarios the predictions of every method are worse than simply returning the target-domain mean. Scenario by scenario, the numbers with a coefficient of determination greater than zero are only: PLSR 11/602, SVR 5/602, CNN 2/572, CNN+MMD 2/568 and Phys+BL 13/568—so even the best-performing Phys+BL genuinely learns something useful in only 2.3% of scenarios. This proportion is the backdrop to all the conclusions of this paper: on these data, cross-origin transfer is a problem that no method has yet solved, and the relative merits of the methods are compared on top of a shared failure. It should be pointed out that the absolute bias of Phys+BL (0.15 °Brix) is the smallest among all methods, so its uncorrected predictions are already the closest to unbiased; this point is of critical significance

Table 6

Aggregate performance of five methods on cross-origin transfer scenarios (R^2 is the across-scenario median; the other columns are across-scenario means)

Method	RMSE/(°Brix)	95% CI	R^2	RPD	Bias/(°Brix)
PLSR	3.321	[3.119, 3.514]	-0.880	0.652	-0.21
SVR	2.777	[2.725, 2.830]	-1.938	0.603	-0.85
CNN	2.109	[2.075, 2.142]	-0.627	0.772	-0.60
CNN+MMD	2.118	[2.089, 2.148]	-0.624	0.766	-0.73
Phys+BL	1.873	[1.784, 1.988]	-0.129	0.913	-0.15

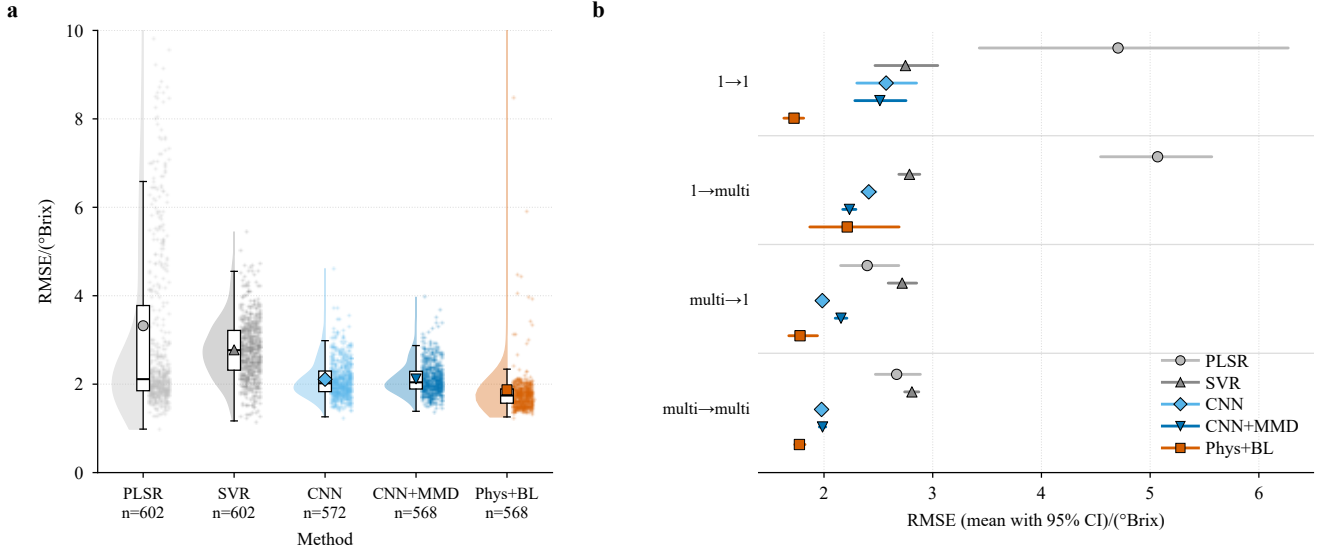


Figure 7: Performance of the five methods under cross-origin transfer. a. RMSE distribution over all scenarios; b. Mean RMSE with 95% confidence intervals, stratified by the four transfer types

in the calibration-transfer comparison of Section 3.7. CNN+MMD (2.118 °Brix) is marginally worse than the CNN, indicating that purely statistical distribution alignment brings no gain on this task. Broken down by transfer type (Fig. 7), the absolute improvement is most pronounced in single-source scenarios (1→1 and 1→multiple) and narrows in multiple-source scenarios, consistent with the expectation that a mixed multiple-source domain has already partly compensated the baseline model.

3.6 Apple case study: ablation of the net contribution of the Beer–Lambert reconstruction loss

To establish whether the improvement of Phys+BL comes from the factorisation architecture or from the BL reconstruction loss, two ablations were run over the five fixed seeds, each scenario’s RMSE being averaged across seeds before comparison (Table 7).

Ablation one (BL loss imposed on a generic CNN): a shallow decoder was added after the CNN feature extractor to reconstruct the spectrum, and a log-domain reconstruction loss of the same form as in Phys+BL was imposed, giving CNN+BL. Across the five fixed seeds, CNN+BL is marginally *worse* than

the CNN (median RMSE 2.112 vs 2.090 °Brix; median paired improvement -1.11%, mean -3.42%), a small but now statistically resolvable degradation (two-sided Wilcoxon $P = 1.1 \times 10^{-2}$, $n=572$); neither variant ever diverges. The BL reconstruction constraint thus brings no benefit to the latent space of a generic CNN, and if anything slightly harms it.

Ablation two (net contribution of the BL loss within the physics architecture): over the 572 scenarios in which both Phys and Phys+BL trained, only the weight of the BL loss was switched. This ablation separates two distinct effects. On *accuracy* the effect is negligible: where both converge, the physics architecture without the BL loss (Phys, median RMSE 1.818 °Brix) performs on a par with Phys+BL (median 1.758 °Brix; median paired improvement +0.63%), a difference tiny in magnitude though resolvable across five seeds ($P = 2.7 \times 10^{-12}$). On *training stability*, however, the effect is clear: defining a test-set RMSE greater than 10 °Brix as “divergence”, the plain physics model diverges in 2.26% of scenario–seed runs (63 of 2792 cells) whereas Phys+BL diverges in only 0.24% (7 of 2858), an exact McNemar test over the discordant runs giving $P < 10^{-10}$. The BL reconstruction loss therefore *does* act as

The label economics of calibration transfer

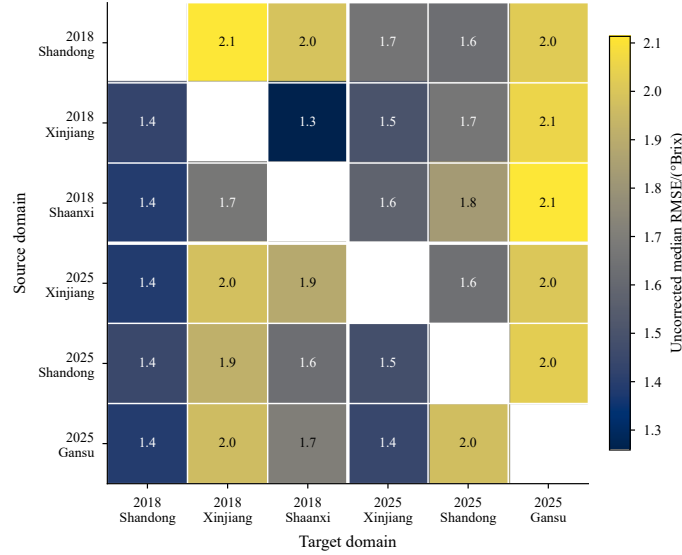


Figure 8: Uncorrected difficulty structure over one-source→one-target cross-origin scenarios: median RMSE of the physics factorisation architecture Phys+BL across the six origin-year domains

Note: rows are source domains and columns target domains; the diagonal (within-domain) is blank, there being no transfer; the white divider separates the 2018 and 2025 seasons; values are means across the five seeds for that one-source→one-target scenario.

a training stabiliser within the physics architecture—reducing divergence roughly ninefold—even though it barely affects accuracy on the runs that do converge.

This stabilising effect is robust to data cleaning, but a single seed can hide it. On uncleaned data the same ablation gave Phys 15/566 and Phys+BL 0/566 divergent scenarios (McNemar $P = 6.1 \times 10^{-5}$); an earlier single-seed analysis (seed 42) on cleaned data then found only 1/552 versus 2/552 ($P = 1.00$) and appeared to show the effect vanishing after cleaning. The five-seed replication resolves the discrepancy: the plain physics model’s divergence rate on cleaned data is still $\sim 2.3\%$ (close to the uncleaned rate), so data cleaning removed the anomalous-refractometer samples (Section 2.4) but not the physics architecture’s intrinsic optimisation instability, and seed 42 simply happened to be a run on which the plain model did not diverge. This is a concrete illustration of why single-seed ablations are unsafe and why the ablation was re-run over five seeds.

Taken together, the five-seed ablations separate two effects of the physical prior. Its cross-origin *accuracy* advantage over a generic CNN (Section 3.5) is carried by the low-rank additive factorisation *architecture*: the explicit BL reconstruction loss adds no accuracy, slightly degrading a generic CNN and leaving the physics architecture essentially unchanged. Its *training stability*, however, comes from the explicit BL loss, which reduces the physics model’s divergence rate roughly ninefold. The physical prior thus has a real but narrow and clearly delimited role—stabilising optimisation of the factorised latent space rather than improving accuracy—which both sharpens and, on

the stability question, corrects an earlier single-seed reading that had missed the effect.

3.7 Apple case study: a calibration-transfer comparison under a matched target-domain label budget

All the comparisons above were made without any post-processing. Slope/bias correction, however, is a classical means of calibration transfer and is applicable to the predictions of any model; Fan et al.[55] showed long ago that applying this correction to apple SSC with only 10 target-domain samples significantly improves cross-batch prediction accuracy. We make no claim as to the novelty of the correction method itself; the contribution here is to apply, for the first time under a unified data split, the same correction symmetrically to a physics-informed architecture and to classical chemometric baselines, thereby testing the previously unanswered question of whether the benefit of a physical inductive bias still holds when a small number of target-domain labels is permitted. The correction was accordingly applied symmetrically to PLSR, SVR, CNN and Phys+BL under a unified split (Table 8).

The results reveal a clear and important pattern. When uncorrected, Phys+BL at 1.873°Brix is significantly better than the CNN at 2.109°Brix (an improvement of 11.3%; the cluster-robust 95% confidence interval excludes zero), consistent with the conclusion over 568 scenarios in Section 3.5. After correction, however, partial least squares regression, support vector regression and the two convolutional networks converge to almost the same level ($1.59\text{--}1.61^\circ\text{Brix}$, Fig. 9a), while Phys+BL becomes the worst

Table 7

Paired comparison of the Beer–Lambert reconstruction constraint imposed on a generic convolutional network versus on the physics factorisation architecture

Comparison	Median RMSE (from)	Median RMSE (to)	Median impr./%	Wilcoxon P	Diverg. (from/to)	n
CNN+BL vs CNN	2.090	2.112	−1.11	1.1×10^{-2}	0%/0%	572
Phys+BL vs Phys [†]	1.818	1.758	+0.63	2.7×10^{-12}	2.26%/0.24%	572
Phys+BL vs CNN	2.090	1.741	+16.81	5.1×10^{-73}	0%/0%	572

Note: all three comparisons use the five fixed seeds [20060515, 20041210, 19810915, 2023, 2024]; each scenario's RMSE is first averaged across seeds and the median over scenarios is reported. [†] Phys+BL vs Phys is the clean loss ablation (identical architecture, only the BL reconstruction loss toggled, script 20); the first and third rows come from script 19, in which no variant diverges. “Median impr.” is the median of the per-scenario relative RMSE reductions (positive = the “to” method is better); mean improvements are omitted, being dominated by the few divergent scenarios. “Diverg.” is the fraction of scenario–seed runs with test-set RMSE $> 10^\circ\text{Brix}$; for Phys vs Phys+BL the exact McNemar test over the discordant runs gives $P < 10^{-10}$. The convolutional backbone in these ablations has a smaller regression head (hidden 16, with dropout) than the CNN of Table 6 (hidden 32), so its RMSE is not directly comparable across tables; within each ablation the architectures are matched.

of all (1.851°Brix): 15.5% worse than CNN+correction (nominal $P = 1.2 \times 10^{-24}$) and 16.7% worse than the simplest PLSR+correction (nominal $P = 2.0 \times 10^{-37}$); the 95% confidence intervals of both paired differences never span zero under scenario, target-set or source-set clustering (see Section 3.5), so the direction is robust.

The difference in the gain each method obtains from the correction reveals the reason for this reversal (Fig. 9b): the gain is as much as 52.2% for PLSR and 24.1% for the CNN, but only 1.2% for Phys+BL. The mechanism must be delimited accurately. The slope/bias correction $y = a \cdot \hat{y} + b$ corrects both the scale (slope a) and the offset (intercept b) of the predictions, so the size of the gain depends on how far the optimal linear transformation between the uncorrected predictions and the target-domain reference values departs from the identity, and not on the bias alone. Indeed, the uncorrected bias of PLSR is not in itself large (-0.21°Brix , see Table 6), and its enormous gain comes mainly from correcting a scale mismatch—the dynamic range of its uncorrected predictions is severely compressed (a very large negative R^2), and a single slope correction recovers much of it. The gain of Phys+BL is small because its uncorrected predictions are already close to that optimal linear transformation in both scale and offset, leaving little room for the correction to recover. In other words, the gain contributed by the physics-informed architecture in cross-domain transfer overlaps numerically to a high degree with a single explicit slope/bias correction, and the latter does the job better and more cheaply. The causal strength and scope of this conclusion must be delimited carefully: the overlap in gains is correlational evidence, and it cannot yet be asserted that what Phys+BL learns *is* a linear recalibration (an alternative reading is that its factorisation absorbs non-linear information and thus underfits after correction); settling the direction requires further representation-level diagnostics (such as tests for non-linearity in the corrected residuals, or the affine explainability of Phys+BL outputs from PLSR/CNN outputs). Moreover, this judgement comes

from apple data alone—a single difficult dataset on which *all methods have already failed* ($R^2 < 0$)—so an 11.3% difference between two models with negative R^2 is a comparison within a stratum of failure. What this paper offers on “whether a physical inductive bias can outperform classical calibration transfer” is therefore an answer that is *negative and reproducible under the present case setting*, not a universal conclusion across datasets.

It should further be noted that even after correction the ratio of performance to deviation of the four methods is only 0.963–1.000; by definition RPD is the ratio of the reference standard deviation to the root mean square error, and a value below 1 means that the prediction is less accurate than simply using the target-domain mean. At the same time the quality misclassification rate remains as high as 34.8%–35.7%. This shows that the intrinsic difficulty of direct cross-origin SSC transfer is not removed by post-processing correction.

The label budget at which this correction takes effect is consistent with the unified benchmark. The 20% target-domain validation set corresponds to roughly 30–40 calibration samples per single-origin target (Section 2.6), i.e. the upper end of the $n \in \{20, 40\}$ grid, which is precisely the range over which the unified reckoning finds simple corrections overtaking the deep/physics envelope under cross-origin/year shift (Section 3.1). The apple case study is thus a finer-grained instance of the same budget-dependent crossover, not a separate phenomenon.

4 Discussion

Why simple corrections tend to win across instruments and seasons. A general observation from this study is that, in a unified head-to-head comparison over 126 tasks from five benchmarks, the more expensive deep and physics-informed representations were not found to be better than the simplest classical corrections, and that the relative merits appear more

Table 8

Effect of the post-hoc slope/bias correction on each method under an identical data split

Method	RMSE, uncorr.	RMSE, corr.	Gain/%	RPD, corr.	R^2 , corr.	Misclass./%
PLSR	3.321	1.589	+52.2	1.000	−0.021	34.8
SVR	2.777	1.597	+42.5	0.992	−0.025	35.0
CNN	2.109	1.602	+24.1	0.986	−0.027	34.9
Phys+BL	1.873	1.851	+1.2	0.963	−0.046	35.7

Note: all cross-origin scenarios were repeated under five fixed random seeds; the across-seed mean was taken first for each scenario, and the scenario was then used as the statistical unit. The numbers of effectively converged scenarios are 602 for PLSR/SVR, 572 for CNN and 568 for Phys+BL (Section 2.6); after correction, Phys+BL+SB diverges numerically in 5 scenarios. “Gain” is the percentage reduction in RMSE after the slope/bias correction relative to the uncorrected value; the paired significance tests are given in the main text of Section 3.7.

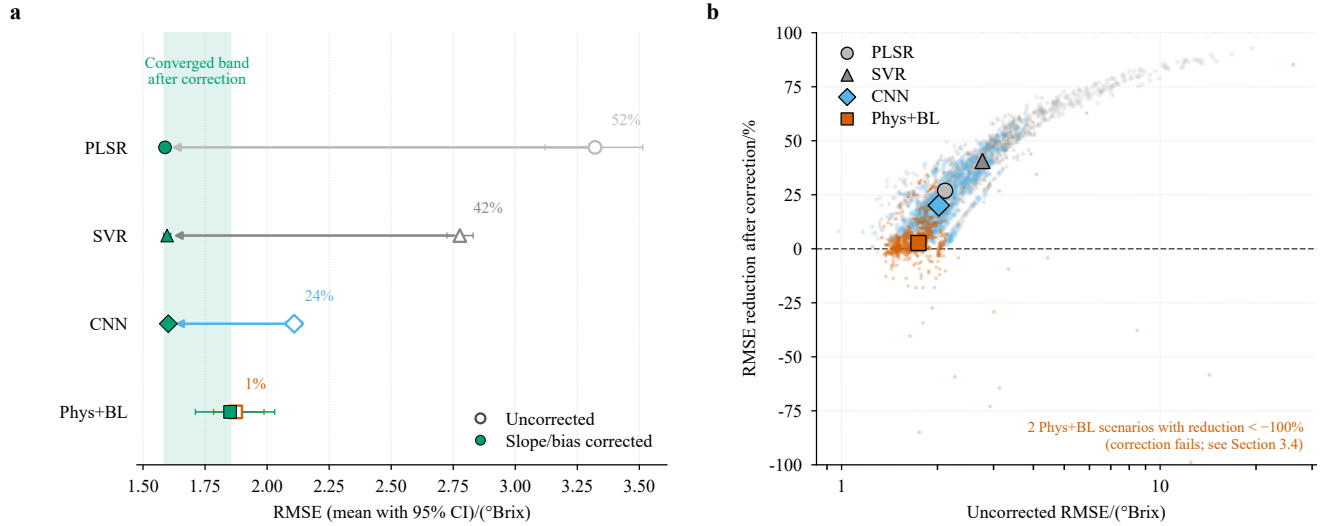


Figure 9: Effect of the post-hoc slope/bias correction on each method. a. Shift in RMSE from the uncorrected (open symbols) to the corrected (filled symbols) state; error bars are 95% confidence intervals from a scenario-level cluster bootstrap, and the shaded band marks the range to which all methods converge after correction. b. Relationship between the gain from correction and the uncorrected RMSE (small symbols: individual scenarios; large symbols: method-wise medians)

consistently associated with the *benchmark and its shift label* than with the continuous magnitude of the shift (Table 4; because shift label is nearly collinear with dataset in the present design, this association cannot be attributed to a main effect of mechanism—see the limitations below). For the instrument and seasonal benchmarks, the simple methods lead in effect size and paired tests at almost every label budget (Cliff’s δ 0.44–0.67, Wilcoxon $P \leq 0.02$). A plausible reason is that inter-instrument differences and whole-season displacements are, in these benchmarks, largely well approximated by *low-order (first- and second-moment) linear transformations* of feature space—translations and rescalings of gain, baseline and scale—which is precisely the structure targeted by second-order moment alignment (CORAL) and slope/bias correction; a deep network, trained on source-domain labels, has discriminative features that are not naturally invariant to such low-order shifts in the target domain and may instead overfit higher-order source-domain features and produce negative transfer. Cross-origin/year shift presents an honest mixture: the deep

methods lead by a slim margin only at near-zero labels (median difference -0.05 to -0.13 , $\delta = -0.09$) and are overtaken by the simple methods beyond 5–10 labels; on cross-laboratory mid-infrared soil the two families are statistically indistinguishable ($P = 0.30$ – 0.62). The practical implication of this picture is clear: under the protocol and label budgets evaluated here, deploying deep domain adaptation by default lacks empirical support, and method selection is better started from the simplest applicable correction than from a deep model.

The feasible boundary of label-free alignment and an in-sample look-up table. Label-free second-order moment alignment reaches the commonly used quantitative screening threshold under pure instrument shift without any target labels (median RPD 0.61→2.01, reaching 2.18 on corn; here $RPD \geq 2$ is an empirical screening-level threshold rather than industrial-grading accuracy, and is a benchmark-level overall median rather than a task-level guarantee), consistent with the physical intuition that instrument shift is close

to a global linear transformation. For seasonal, cross-laboratory and cross-origin shift, however, the same alignment achieves only a partial recovery (RPD rising to 1.02–1.27), indicating that these mechanisms contain non-linear or higher-order components that second-order moment alignment cannot capture. Organising “shift type $\times$ label budget $\rightarrow$ most frequent best method” into an in-sample frequency map yields a heuristic starting point rather than a validated rule: at zero labels, prefer label-free alignment; as the budget grows, instrument shift turns to model updating and population/seasonal shift to slope/bias correction, until a target-only model catches up when labels are plentiful. Because it is fitted in-sample on these five benchmarks, this map is a hypothesis-generating summary that requires prospective validation before operational use.

No unified transferability scaling law was observed under the present design. One natural expectation is that, since shift is measurable (distribution distances, intrinsic dimensionality), the crossover budget n^* should be predictable from these quantities, thereby forming a “transferability scaling law”. We tested this directly on 126 tasks from five benchmarks and give a *negative* answer (within the present design): the log-log fits of n^* on measures of domain shift have R^2 of only 0.002–0.25, leave-one-benchmark-out extrapolation reaches a mean absolute error of 3.6–34 labels on the original scale, and conformal interval coverage is erratic. One possible root cause is that domain shift across mechanisms is nearly incommensurable under the metrics used here—the same numerical MMD corresponds to widely different transfer difficulties under the four mechanisms of instrument, season, origin and laboratory, and is hard to characterise by a single continuous scalar. What can be stably reproduced is therefore only the ranking of methods over *discrete* shift mechanisms (the look-up table), not a quantitative extrapolation from continuous shift magnitude. This negative observation has methodological value in itself: it suggests that the idea of predicting transfer difficulty data-drivenly from marginal distribution distances does not work in this setting, and that one should return to mechanism classification. Its strength, however, must be strictly delimited (see below).

The boundaries of identifiability and external validity. Four limitations must be declared honestly; together they constrain how far the conclusions above generalise. First, and most critically, **shift mechanism and dataset are nearly collinear**: seasonal shift is carried only by mango, cross-laboratory shift only by OSSL soil and cross-origin/year shift only by our in-house apples, the number of independent benchmarks is in fact $n = 5$, and the 126 transfer tasks are highly correlated within benchmarks (sharing parent samples, instruments and preprocessing). “Performance differences are determined by the *mechanism* of the

shift” and “...by the *dataset*” therefore cannot be fully separated in this design, and the win rates and significance levels of the paired tests may be optimistic because of pseudo-replication within benchmarks; the mechanism-level conclusions of this paper should be read as an *exploratory empirical map* rather than as confirmation of a main effect of mechanism, and a strict test would require at least two independent benchmarks per mechanism. Second, **taking the “best method within a family” on each task** introduces a selection bias (winner’s curse), compounded by the unequal family sizes (two simple, four deep); the envelope of Table 4 is therefore reported as a descriptive upper bound. To guard against this, we also report a selection-free comparison (Table 5)—a pre-specified representative pair and the family mean—which preserves the pattern and, at near-zero labels under cross-origin shift, mildly favours the simple methods where the envelope had mildly favoured the deep family; the “simple is at least as good” reading does not appear to be an artefact of the envelope selection. Third, **label-free alignment is transductive**: CORAL uses the second-order moments of unlabelled target-domain spectra, an acceptable form of transductive adaptation that must nevertheless be declared explicitly, and its “zero-label usability” is not equivalent to inductive deployment on entirely new samples. Fourth, **the tuning budget of the deep baselines is limited**, so on small-sample tasks (such as apple at 0–5 labels) their disadvantage may stem partly from data starvation rather than from a structural trade-off; “simple wins” should be understood as holding “under the matched compute and tuning budget of this study”, not as an upper bound on the capacity of deep methods. We note additionally that the qualitative pattern is not driven by the in-house apple data, which supply 72 of the 126 tasks: restricting attention to the 54 tasks of the four *public* benchmarks leaves the reading unchanged—simple methods are better under instrument and seasonal shift and indistinguishable on cross-laboratory soil—so the only mixed case is the apple (cross-origin/year) benchmark. Finally, some methodological choices bound how far these results should be pushed. The main tables report native-unit RMSEP for interpretability; we verified that the conclusions are unchanged on the dimensionless NRMSEP = 1/RPD and at the benchmark level of analysis (Section 3.1, script 68), but a fully hierarchical benchmark $\rightarrow$ domain-pair $\rightarrow$ task error model is still limited by the small number of benchmarks ($n = 5$), so benchmark-level inference is low-powered and we report it as directional. PDS, moreover, is a standardisation designed for paired transfer standards; it is informative mainly on the paired-instrument benchmarks and should be read with caution on the unpaired population shifts. These points do not alter the pragmatic conclusion that deep domain adaptation does not warrant default deployment on

evidence of this kind, but they do require the headline findings to be stated as *bounded empirical findings* rather than universal laws—which is why this work is positioned as an honest reckoning rather than as the proposal of a purportedly universal new algorithm.

We now turn to the in-depth case study of cross-origin transfer on our in-house apple data, focusing on one concrete and pointed question: can a physics-informed architecture that encodes the Beer–Lambert law as an inductive bias outperform classical calibration transfer?

A physico-chemical explanation for the destabilisation of BL-type additive structure. Intact fruit is a turbid medium in which photon transport is governed jointly by absorption and reduced scattering[56]; cross-origin differences in soil minerals, irrigation and accumulated temperature systematically modulate the histological and biochemical quantities that determine the reduced scattering coefficient (pectin and cellulose volume fractions, intercellular air, wax thickness, starch granule density), changing the spectral shape and magnitude of the scattering coefficient and thereby rescaling the effective path length in a wavelength-dependent manner; a chromophore structure that was originally additive becomes multiplicative across bands, destroying the low-rank factorisation learned on the source domain—which is precisely the mechanism that manifests as BL residual amplification. The two largest amplification ratios (Xinjiang→Shaanxi 4.55 and Xinjiang→Shandong 3.93) occur at the origin pairs that would, a posteriori, be expected to differ most in soil and microclimate, and the smallest ratio between two geographically close eastern origins; we did not measure these environmental variables directly, however, so this correspondence is offered as a post-hoc interpretation rather than a tested relationship. This mechanism also explains why pure scatter correction cannot fully remove cross-origin shift—a single scalar rescaling cannot eliminate a wavelength-dependent scattering drift.

Why the factorisation architecture, more than the physical loss, may carry the advantage. The latent features of a generic one-dimensional convolutional network are data-driven spectral features without a BL additive component structure; imposing a BL reconstruction loss makes the network oscillate between two conflicting gradients—reconstruction fidelity (requiring decomposable features) and minimisation of prediction error (requiring discriminative features)—and the prediction loss ultimately suppresses the physical constraint, yielding a suboptimal solution. The physics architecture presets the latent space to a three-component structure, so that the physical constraint and the inductive bias are aligned and reinforce each other. It must be stressed that our latent factorisation is not identifiable at the molecular level: the components the network learns should not be interpreted

as molar absorptivities or concentrations of specific chromophores, they are merely the structural carriers that make a BL-type additive constraint imposable. We therefore make no chemical assignment for the latent components, and the value of the physical prior lies in the limited accuracy advantage of its factorisation architecture when uncorrected and in label-free risk warning, not in training stability or chemical interpretability.

Interpreting negative R^2 and RPD below 1.0. Under the standard held-out definition, a negative R^2 means the predictions are worse, in squared-error terms, than simply predicting the target-domain mean; it is a genuine indication of poor absolute fit, not an artefact to be waved away. It is nonetheless common in cross-domain evaluation, because the target-domain mean is not available in advance during deployment, so a source-only model can carry useful relative (rank) information while still losing to the retrospective target mean. At this level of difficulty we therefore compare methods by relative improvement together with the median and the bias, which are more robust to a few divergent scenarios, while being explicit that the absolute fit is poor. It must be frankly conceded that the absolute performance reported here is markedly lower than that of comparable studies in the literature. This gap arises mainly from the severity of our setting—zero target-domain labels, direct cross-origin transfer, and no calibration transfer at all in the main experiments. By way of contrast, Fan et al.[55], in a long-term evaluation of apple SSC, applied a slope/bias correction with only 10 target-domain samples and reduced the prediction root mean square error from 0.704%–1.716% to 0.501%–0.654%, far better in absolute terms than any method here. A quality misclassification rate still around 40% and a still-negative median R^2 remain markedly short of the accuracy attainable in single-origin deployment in that literature (for example the 0.501–0.654 reported by Fan et al.). Our method is positioned as risk warning and a robustness improvement for cross-origin transfer, not as a grading model ready for production.

Relation to classical calibration transfer, and engineering recommendations. The most important boundary finding of this study is that the advantage of the physical prior over a generic network not only disappears but reverses once a single post-hoc slope/bias correction is applied (Table 8). The gains each method obtains from the correction (PLSR 52.2%, CNN 24.1%, Phys+BL 1.2%) suggest that the lead of Phys+BL when uncorrected may derive mainly from components that overlap with a linear recalibration (scale and offset correction); once a small number of target-domain labels is allowed to perform that correction explicitly, a complex architecture no longer shows any advantage over the classical correction. The strength of this causal reading is subject to the limitations noted in Section 3.7 (correlational evidence, a single dataset

on which all methods fail). This has direct implications for engineering practice: if a small number of calibration samples (20% in this study) can be acquired at a new origin, the simplest PLSR with a slope/bias correction already achieves the best cost-effectiveness and there is no need to deploy a deep model[18, 22]; the value of the physical prior should be located in three places—the limited accuracy advantage of its factorisation architecture over a generic network when uncorrected and when target-domain labels are extremely scarce; *training stability*, a five-seed ablation (Table 7) showing that the BL reconstruction loss cuts the plain physics model’s divergence rate roughly ninefold (2.3% to 0.2% of scenario-seed runs, McNemar $P < 10^{-10}$)—a robust benefit that an earlier single-seed analysis had missed; and label-free risk warning based on the BL residual (BL residual amplification correlates significantly with transfer risk and can provide a preliminary reference for cross-origin deployment before any reference label is acquired). What the reconstruction loss does *not* buy is accuracy—it adds nothing on the runs that converge and slightly degrades a generic CNN—so the accuracy advantage is carried by the architecture and the stability advantage by the explicit constraint.

Other limitations and outlook. The BL loss weight is scenario-dependent, and we recommend tuning it quickly on a small calibration set; $K=3$ is an engineering choice, and a learnable number of components could be explored; architecture-constraint compatibility has so far been verified on only one CNN backbone, and verification on multiple backbones (ResNet1D, Transformer1D) is left to future work; the cross-instrument variation in this study is confined to two generations of spectrometer from the same manufacturer (one in each of 2018 and 2025, aligned to 229 common channels) and does not cover instrument differences between manufacturers or between optical layouts, which are precisely the main battleground of the calibration-transfer literature[22, 39]; whether the BL-inspired residual has incremental value over generic domain-distance metrics (such as principal-component reconstruction distance or maximum mean discrepancy) in predicting cross-domain risk has not been directly compared here and is left to future work; and direct measurement of the reduced scattering coefficient (for example by spatially resolved diffuse reflectance) would upgrade our risk characterisation from a data-driven surrogate to a rigorous physical quantification.

5 Conclusions

Across five calibration-transfer benchmarks, 126 transfer tasks, four types of domain shift and two spectral modalities, the simplest classical corrections were, on these benchmarks, at least as good as deep and physics-informed representations. Label-free second-order moment alignment and the two-parameter slope/bias

correction led at almost every label budget under instrument and seasonal shift (Cliff’s δ 0.44–0.67, Wilcoxon $P \leq 0.02$); under cross-origin/year shift the deep envelope held a slim edge only at near-zero labels and was soon overtaken (a margin that does not survive a selection-free reanalysis, Table 5); and on cross-laboratory mid-infrared soil the two families were statistically indistinguishable. Relative merit was more consistently associated with the benchmark and its shift label than with the continuous magnitude of the shift, and transductive label-free alignment reached a screening-level threshold under pure instrument shift from unlabelled target spectra alone (median RPD 0.61→2.01). We summarise this as an in-sample frequency map over shift type and label budget—requiring prospective validation, since shift label is near-collinear with dataset ($n = 5$)—whose pragmatic message is that deep domain adaptation does not warrant default deployment here.

Correspondingly, the crossover budget n^* at which a simple method catches up could not be predicted from continuous measures of domain shift: log-log fits reached R^2 of only 0.002–0.25, and leave-one-benchmark-out extrapolation failed, with a mean absolute error of 3.6–34 labels and erratic conformal coverage. Within the metrics and benchmarks examined, transfer difficulty is not captured by a single scalar; what reproduces stably is the ranking of methods over *discrete* mechanisms, not extrapolation from shift magnitude.

The apple case study delimits the physical prior. Beer–Lambert additive low-rank structure destabilised systematically across origins (residual amplification 1.08–4.55-fold, all 12 scenarios $P < 0.05$) and correlated with transfer risk (median Spearman $\rho = 0.43$), making it usable as a label-free risk warning before any target label is acquired. The factorisation architecture was the best uncorrected method among methods that all fail in absolute terms (1.873 °Brix, 11.3% below a matched-budget convolutional network); a five-seed ablation shows that this accuracy advantage comes from the low-rank additive factorisation architecture—the Beer–Lambert reconstruction loss adds no accuracy and slightly degrades a generic CNN—whereas the loss itself acts as a training stabiliser, cutting the physics model’s divergence rate roughly ninefold. Once 20% of target labels were spent instead on a slope/bias correction, the classical baselines converged to 1.59–1.61 °Brix and overtook it.

These findings are bounded. Each shift mechanism is carried mainly by a single dataset ($n = 5$ benchmarks), so mechanism and dataset cannot be fully separated, and the negative scaling result states what was *not observed under this design* rather than what does not exist. On the apple data every method fails in absolute terms—even after correction the RPD remains below 1 and the quality misclassification rate

34.8%–35.7%—so partial least squares regression with a slope/bias correction is the most economical option available, not a production-ready one.

CRedit authorship contribution statement

Panlin Li: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Writing – original draft. **Yuchang Li:** Investigation, Validation, Data curation. **Longjie Li:** Software, Investigation, Data curation. **Ya Liu:** Resources, Investigation, Data curation. **Yutong Feng:** Software, Validation. **Muhammad Waqar Akram:** Writing – review & editing. **Junxian Guo:** Supervision, Methodology, Resources, Writing – review & editing. **Hua Huang:** Conceptualization, Supervision, Project administration, Funding acquisition, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The four public benchmarks (Cargill corn, IDRC 2002 tablet shoot-out, OSSL global soil spectral library, and the Anderson mango dataset) are available from their original sources cited in Table 2 and are not re-hosted here. The apple near-infrared spectra are internal data of Xinjiang Agricultural University and have not been publicly released; researchers who wish to reproduce or extend the experiments may request access from the corresponding author on a case-by-case basis, subject to the data-sharing policies of the university. The analysis code is available at <https://github.com/2004lryan/015calibration-transfer-label-economics>.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to assist with English-language polishing (translation and editing) of author-written drafts. The tool was not used to generate or interpret experimental data or to formulate the scientific claims. After using this tool the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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